

Don't Judge Citizens for Changing Their Opinions¹

Nicholas C. Dias

Abstract

Citizens often change their minds about policy. Scholars take this instability as evidence that citizens' opinions are not "meaningful"—a failure of competence that leaves democracy without preferences to represent. Yet scholars rarely define the term, and direct tests of whether stability indicates meaningfulness are almost nonexistent. This article addresses both gaps. Drawing on a long tradition in public opinion research, I define an opinion as meaningful when it is consistent with a citizen's fundamental goals, or "values." Using panels from three countries ($n > 29,000$), I show that value-consistent opinions are only slightly more stable than value-inconsistent ones. Consequently, stable opinions are, at most, five percentage points more likely to be value-consistent. These findings suggest that a dominant indictment of citizen competence and democratic representation—that opinion instability reveals an absence of meaningful opinions—rests on an empirical premise that, in its first large-scale test, finds little support.

Keywords: opinion stability; values; citizen competence; democratic representation; panel data

Short Title: Opinion Stability and Value-Consistency

Word Count: 7,600

¹Claude (Anthropic; models `claude-fable-5-1` and `claude-opus-5-5`), accessed via the Claude Code application between July 26 and October 6, 2026, was used to copyedit this article, to debug the replication code, and to format the manuscript for submission. All output was reviewed and edited by the author.

Can citizens’ policy opinions offer meaningful guidance to policymakers? Many political scientists fear that the answer is no. While citizens are often willing to voice opinions about public policies, these opinions frequently change over time. Since Converse (1964), scholars have read such instability as evidence that citizens’ opinions are superficial or, at least, unreliable indicators of what citizens actually value (Zaller 1992). Other work makes the reverse inference, treating stable opinions as meaningful (Ansolabehere, Rodden, and Snyder 2008; Vivyan, Lauderdale, and Hanretty 2026). And because most citizens’ opinions are unstable on most issues, the implication is that most policy opinions are meaningless.

This argument casts a shadow over citizen competence and representation, which presuppose that citizens have meaningful opinions to which governments can and should respond. Citizens vote directly on consequential policy referendums (e.g., Brexit), and their policy opinions strongly shape the candidates they support (Ansolabehere and Kuriwaki 2022; Mummolo, Peterson, and Westwood 2021). If those opinions do not reflect what citizens actually value, citizens risk voting for policies at odds with their own goals (Lau and Redlawsk 1997), elected officials are left without guidance on what policies to implement (Butler and Nickerson 2011), and elections cannot discipline governments to serve citizens’ ends (Achen and Bartels 2017). As Achen (1975) warned, if citizens lack meaningful policy opinions, “[d]emocratic theory loses its starting point” (1220).

Yet the idea that an opinion’s stability indicates its meaningfulness rests on two claims that have gone largely unexamined. The first is conceptual: that “meaningful” refers to some definite property of opinions that democratic representation requires. However, scholars have defined the term rarely and inconsistently. The second is empirical: that stability actually tracks this property. Almost no study has tested this assumption directly. Unless both claims can be substantiated, opinion instability cannot establish that citizens are incompetent or that representation lacks its starting point. This article examines both claims.

First, I propose an explicit, minimalist definition of a “meaningful” opinion. Drawing on a deep tradition in public opinion research, I treat a policy opinion as meaningful when it is

consistent with the fundamental goals that guide citizens’ judgments—what I call “values” (Rokeach 1973; Schwartz et al. 2012). This is the form of “meaningfulness” most relevant to representation: An opinion cannot steer government toward what a citizen wants if it points away from those goals.

Second, I test whether value-consistent opinions are more stable over time. Within panel datasets from Germany, the Netherlands, and the United States ($n > 29,000$), I identify pairs of basic values and policy issues wherein prior literature consistently ties the value (e.g., concern for others) to a particular opinion (e.g., support for welfare). Value-consistent opinions, I find, change nearly as much as value-inconsistent ones. Knowing an opinion is stable raises the probability that it is value-consistent by no more than five percentage points.

If stability cannot discriminate between value-consistent and -inconsistent opinions, then decades of research on opinion stability tell us less about citizen competence and democratic representation than is often assumed. My results also suggest we need new tools for evaluating the competence of citizens’ opinions. Rather than asking how often citizens’ opinions change, we should ask what changes them and why.

Why Worry About Opinion Stability?

For six decades, political scientists have wrestled with a seemingly simple question: How often do citizens change their minds about policy issues? This debate began with Converse (1964), who provocatively suggested that all but a “hard core” of the public hold meaningless opinions that vacillate over time (242–243). Achen (1975) responded that much of the apparent change reflects measurement error from ambiguous survey questions. Wave after wave of researchers—armed with better data, newer models, and softer assumptions—has since tried to disentangle true opinion change from statistical noise, each arriving at a different balance (Ansolabehere, Rodden, and Snyder 2008; Erikson 1979; Feldman 1989; Lauderdale, Hanretty, and Vivyan 2018).

But why has opinion stability received so much attention? It is not because stability is important in itself: Stability matters because it is taken to signal whether citizens’ policy opinions are meaningful.²

The connection between stability and “meaningfulness” has been read in two ways. The first reading treats instability as evidence against meaningful opinion. Converse (1964) argued that a fluctuating policy opinion is “*prima facie* evidence” (241) that the opinion is unconstrained by “some superordinate value or posture toward man and society” (211). Unstable opinions, in this reading, represent non-attitudes invented on the spot to satisfy researchers (Converse 1964) or clumsy amalgamations of whatever considerations happened to be top-of-mind when a question was asked (Zaller 1992). Because ordinary citizens’ opinions lack such constraint, Converse (1964) claimed, only 65% manage to “locate themselves even on the same *side* of the controversy” after two years. By contrast, he expected that 90% of legislators, whose opinions presumably are so constrained, would hold the same stance (239).

The second reading, more common in later work, treats stability as evidence for meaningful opinion. As Vivyan, Lauderdale, and Hanretty (2026) write, “a respondent is considered more likely to have a well-formed opinion on an issue to the extent that their stated opinion on that issue remains consistent” across panel waves (23). Similarly, Freeder, Lenz, and Turney (2019) ask “[w]hat share of citizens hold meaningful views about public policy?” (274) and answer by counting stable opinions (288–289).

Thus, scholars’ fixation on opinion stability is animated by concerns about citizen competence and, ultimately, democratic representation (Price and Neijens 1997). The benchmark

²That said, stability may matter for political influence: An opinion can shape behavior only insofar as it persists (Luttrell and Togans 2021), and durable opinions may be easier to represent (Krosnick 1988, 253). Yet aggregate support for a policy—the quantity policymakers presumably attend to—is usually stable even when individual opinions are not (Page and Shapiro 1992). Moreover, we should want citizens to change their opinions when confronted with relevant information (Delli Carpini and Keeter 1996, 234–235).

of empirical research on representation is “the continuing responsiveness of the government to the preferences of its citizens” (Dahl 1971, 1)—a standard that presupposes citizens hold meaningful preferences to which government can respond (Sabl 2015). Pervasive opinion instability seems to dissolve the presupposition: If citizens’ survey responses wander unpredictably, there are seemingly no preferences for elected officials to represent and no fixed standard by which voters can hold politicians accountable (Freder, Lenz, and Turney 2019). As Inglehart (1985) put it, “[I]f most citizens don’t really *have* any coherent or stable preferences about major political issues, why should political decision makers take them into account? Indeed, how *could* they?” (97).

The stability debate, in short, has never been about stability for its own sake: Stability is used as a proxy for meaningfulness. Whether it can bear this weight depends on the two claims identified above: that “meaningful” has a fixed referent, and that stability tracks it. The next two sections take each in turn.

What Is a Meaningful Opinion?

For all its influence, the reasoning above leans on a term—“meaningful” opinions—that the literature defines rarely and inconsistently. The term is used interchangeably with others, such as “real,” “genuine,” “high-quality,” and “sophisticated.” Even work that treats opinion quality systematically tends to enumerate criteria without specifying how they relate. For instance, Price and Neijens (1997) catalog “[s]tability, lack of volatility” and “[c]onsistency with values, other opinions” among many other quality criteria (346), concede that several are contradictory, and propose “no immediate solution to this problem” (355).

This slippage is consequential because the various quality criteria are conceptually and empirically distinct (Conner, Wilding, and Norman 2022; Howe and Krosnick 2017). To some, “meaningful” denotes bare authenticity—i.e., that some real, non-random evaluation underlies the survey response. To others, it encompasses further virtues: The opinion is well-informed

(Delli Carpini and Keeter 1996), coherently connected to other beliefs (Converse 1964), and resistant to arbitrary influence (Druckman 2001). This leaves unclear what characteristic stability is supposed to proxy for.

Therefore, this article fixes a referent. Common to scholars’ varied uses of “meaningful” is a concern about democratic representation. I thus use the term in a narrow, representational sense: A policy opinion is representationally meaningful when it is consistent with the citizen’s fundamental goals—what I will call “values” (Rokeach 1973; Schwartz et al. 2012).³ Values specify the ends that citizens wish to achieve, while public policies are possible means to those ends. A representationally meaningful opinion, then, selects the policy that better advances a citizen’s ends. Citizens can steer governments toward such policies only if their expressed opinions point toward, rather than away from, their values (Ansolabehere, Rodden, and Snyder 2008; Mummolo, Peterson, and Westwood 2021).

This definition has a long history at the center of public opinion research. Converse (1964) himself was interested in identifying opinions constrained by values, and the most influential operationalizations of opinion quality are largely derivative of value-consistency (Sniderman and Bullock 2004, 337). For instance, the expectation that policy opinions should correlate with one another is only sensible if these opinions flow from the same superordinate value (Peffley and Hurwitz 1985). Likewise, political knowledge matters in large part because, without it, citizens cannot “connect their values with concrete matters of politics” (Delli Carpini and Keeter 1996, 229).

Most importantly, meaningful opinions are expected to persist because values persist. Values represent a citizen’s fundamental goals, and they remain stable over time (Milfont, Milojev, and Sibley 2016; Vecchione et al. 2020; Zaller 1992). A policy opinion is a judgment about whether a policy advances those goals. When the two agree, the opinion has something

³The definition of “values” is deliberately broad: A value can be a moral commitment, such as concern for the welfare of others, but it need not be. In the theory of values introduced below, the self-interested desire to accumulate resources is also a value (Schwartz 2012).

Table 1: Proposed Links Between Value Grounding and Stability

Quote	Source
“Another likely reason for the enhanced stability of important policy attitudes is linkage between these attitudes and core values.”	Krosnick (1990, 68)
“[G]eneral orientations—under most conditions—should be fairly stable. And if these beliefs structure more specific attitudes . . . orientations like militarism may provide the very stabilizing ‘anchors’ to policy attitudes.”	Peffley and Hurwitz (1993, 63)
“[T]he ease with which [citizens] can be blown from one side of an issue to the other suggests that the positions they take are far from securely anchored in underlying, enduring principles.”	Sniderman and Theriault (2004, 134)
“[T]he more people moralize an attitude, the less they tend to change that attitude in response to persuasive arguments.”	Luttrell and Togans (2021, 552)
“[P]eople adopt political attitudes consistent with their ideology and party identity, which in turn holds these political attitudes in place and keeps them stable.”	Turner-Zwinkels and Brandt (2025, 126)

anchoring it in place: The citizen wants what the policy delivers, so any case for abandoning the opinion must overcome a core value. When the two disagree, the opinion works against the citizen’s values, which push the citizen to change their opinion every time the issue comes to mind. A value-consistent opinion, in short, is one a citizen has a strong and durable reason to keep, and a value-inconsistent opinion is one they have a strong and durable reason to give up. Table 1 shows how deeply the connection between value-consistency and stability runs in the literature.

Indeed, the idea that values organize policy opinions has spawned a sizable literature, which finds that political values of many kinds predict policy opinions in their respective domains (for a review, see Feldman 2003). Beliefs about equality and economic individualism structure social welfare opinions (Feldman 1988), general postures like militarism anchor foreign policy attitudes (Peffley and Hurwitz 1993), and moral traditionalism organizes positions on cultural controversies (Goren 2012). More recent work grounds policy opinions

in a small set of basic human values (Enke, Rodríguez-Padilla, and Zimmermann 2023; Goren et al. 2016)—a move I return to below.

To be sure, opinions may also be constrained in other ways—e.g., when citizens adopt the bundle of positions that their political party or another social group endorses (Converse 1964; Elder and O’Brian 2022; Freeder, Lenz, and Turney 2019). Such “social” constraint can also produce opinion stability, but its implications for democratic representation are ambiguous: A socially constrained opinion need not advance the citizen’s own fundamental goals. Indeed, a large literature laments how readily parties sway citizens to endorse or oppose policies, whatever their substance (for a review, see Bullock 2020).

Value-consistency, in short, is the criterion most relevant to the representational conception of “meaningfulness.” All others—stability included—are largely proxies for value-consistency. I hereafter use “value-consistent” and “meaningful” interchangeably.

Does Opinion Stability Signal Value-Consistency?

The remaining question is empirical: Does stability track value-consistency? If stability is a strong proxy for value-consistency, then research on opinion stability doubles as evidence about citizen competence and democratic representation: Counting stable opinions would reveal how many citizens form opinions competently and how many value-consistent preferences there are for government to represent. Whether stability is a strong proxy, however, is the second unexamined claim.

Despite six decades of research connecting value-consistency to opinion stability, direct evidence on whether stability tracks value-consistency is scarce. To my knowledge, only one study has examined the relationship (Peffley and Hurwitz 1993). That study, however, re-interviewed just 301 residents of one U.S. city and tracked only foreign policy attitudes over one year. It also focused on political values, which, as I discuss later, may not be exogenous to citizens’ partisanship or policy opinions (Goren 2005; Goren, Federico, and Kittilson 2009).

The remaining evidence for the consistency-stability link is indirect and mixed. Some studies suggest that politically knowledgeable citizens hold more stable opinions (Converse 2000; Delli Carpini and Keeter 1996; Feldman 1989; Kinder and Kalmoe 2017), while others find little difference (Achen 1975; Ansolabehere, Rodden, and Snyder 2008; Erikson 1979). Even where they do, it is unclear why. Knowledgeable citizens may be more likely to form value-consistent opinions (Zaller 1992). But the pattern is also compatible with motivated reasoning (Taber and Lodge 2006): The informed may simply be better at defending their views, regardless of whether those views align with their values.

A separate line of research links opinion stability to self-reported characteristics that might accompany value-consistent opinions. Krosnick (1990) finds that stability correlates positively with issue importance, which he partly attributes to the “linkage between these attitudes and core values” (68). However, when examining a wider range of panel datasets, Leeper (2014) finds this relationship to be small and statistically unreliable. Similarly, Luttrell and Togans (2021) find a small correlation between stability and moral conviction—i.e., the perception that an opinion reflects one’s notions of right and wrong. Yet moral conviction is distinct from consistency as defined above: People are unreliable reporters of why they hold their opinions (for a review, see Pronin 2009), may misunderstand how their values bear on policy positions (Zaller 1992), or may be influenced by their values without knowing it (Jost 2021). Simply put, an opinion can feel value-based without being value-consistent, and vice versa.

In sum, the assumption that stability can gauge value-consistency rests on thin empirical ground. The remainder of this article tests it directly.

Which Opinions Are Value-Consistent?

Assessing whether opinion stability tracks value-consistency first requires identifying which opinions reflect citizens’ deeper values. The most direct approach compares citizens’ policy

positions with their stated values (for a review, see Feldman 2003).⁴ For instance, scholars might compare citizens’ endorsements of statements such as “It’s very important to you to help the people around you” with their opinions on welfare policies (e.g., Goren et al. 2016).

Importantly, political scientists distinguish between political and basic values. Political values are narrower, representing “principles and belief assumptions about government, citizenship, and American society” (McCann 1997, 565). For example, “militarism” captures whether a citizen prefers an assertive or accommodating military posture and predicts opinions on issues like defense spending (Peffley and Hurwitz 1993). However, because political values closely resemble the opinions they are meant to explain, some scholars question whether they are distinct from, or exogenous to, policy opinions (Arceneaux et al. 2025). Indeed, citizens’ political values are roughly as unstable as their policy opinions (see app. F) and susceptible to elite influence (Goren 2005; Goren, Federico, and Kittilson 2009; Vecchione et al. 2013). Any observed relationship between political values and policy opinions could therefore reflect reverse causation or a common cause like partisanship.

⁴Another popular strategy treats opinions as meaningful when they match citizens’ self-reported ideology—e.g., when a “conservative” opposes universal healthcare (Kinder and Kalmoe 2017). However, many citizens misunderstand ideological labels (Ellis and Stimson 2012), so self-labels might not signal much about underlying values. Moreover, ideological self-labels may function less as value summaries than as group identities: Self-identified conservatives are prone to endorse any policy labeled conservative, regardless of its actual substance, and self-identified progressives do the same (Malka and Lelkes 2010).

Table 2: Higher-Order Values

Focus	Name	Description
Social	Self-Transcendence	Prioritizes the well-being of others, especially those different from oneself, and the natural environment. Expressed through universalism and benevolence.
	Conservation	Emphasizes stability, order, and continuity. Rooted in security, conformity, and tradition, it favors preservation of existing arrangements over rapid change.
Personal	Self-Enhancement	Centers on personal success, status, and control over resources or people. Linked to power and achievement, it advances self-interest.
	Openness to Change	Values independence, novelty, and variety. Grounded in self-direction, stimulation, and hedonism, it encourages exploration and new experiences.

Source: Schwartz et al. (2012).

For these reasons, recent work has focused on basic values (e.g., Goren et al. 2016). The most widely used psychological framework (Schwartz et al. 2012) identifies four higher-order values fundamental to people’s moral outlooks (see Table 2). Of particular interest to political scientists are the two social values: self-transcendence (concern for the welfare of others, especially those different from oneself) and conservation (desire for physical safety, social order, and the preservation of cultural customs). These values deserve special attention because they dictate how one relates socially to others (Schwartz 2012, 13–14), and citizens tend to form policy opinions based on what they think is best for society as a whole (Sears and Funk 1991).⁵

Unlike political values, basic values are plausibly exogenous to both partisanship and policy opinions. They guide people’s attitudes and behaviors across all domains of life (Schwartz et al. 2012) and emerge by age eleven, before most political socialization occurs

⁵Indeed, the effect of “personally focused” values on policy opinions is weaker and less consistent (Goren et al. 2016, 993). Relatedly, self-interest matters mainly when a citizen’s personal stakes are obvious and large (Horowitz and Levendusky 2011)—stakes most citizens lack in many policy issues (for a review, see Sears and Funk 1991).

(Collins et al. 2017; Döring et al. 2015). Finally, they are considerably more stable than policy opinions over long periods (app. G).

Basic values also subsume political values, strongly predicting those studied most in public opinion research—e.g., moral traditionalism, limited government, and equality (Schwartz et al. 2014). They also predict citizens’ policy opinions across many domains: Self-transcendence predicts support for economic redistribution, racial equality, immigration, and civil liberties, while conservation predicts support for strict crime policies and military interventions as well as opposition to immigration, racial equality measures, and progressive social policies like gay marriage (for a review, see app. B). The logic of these pairings is straightforward. The positions that self-transcendence favors extend resources, rights, or protection to everyone, including strangers and the natural world. By contrast, the positions that conservation favors guard against threats to safety or to settled ways of life. These relationships are consistent across political cultures, suggesting that values and policy opinions are naturally aligned. In sum, basic values capture the substance of political value constructs while offering stronger guarantees of exogeneity.

Methods

To test whether value-consistent opinions are more stable over time, I use three panel datasets from Germany, the Netherlands, and the United States (summarized in Table 3).⁶ These

⁶These datasets are the German Longitudinal Election Study, the Longitudinal Internet Studies for the Social Sciences managed by Centerdata (Tilburg University, the Netherlands), and the General Social Survey’s 2008–2012 panel (GLES 2023; Scherpenzeel and Das 2010; Smith, Marsden, and Hout 2018). This study involves only secondary analyses of de-identified data distributed for scientific use. It was conducted in compliance with relevant laws, entailed no interaction with human participants, and, under the regulations applicable at the author’s institution, did not require review by an institutional or national research ethics committee.

Table 3: Panel Datasets Used in Analysis

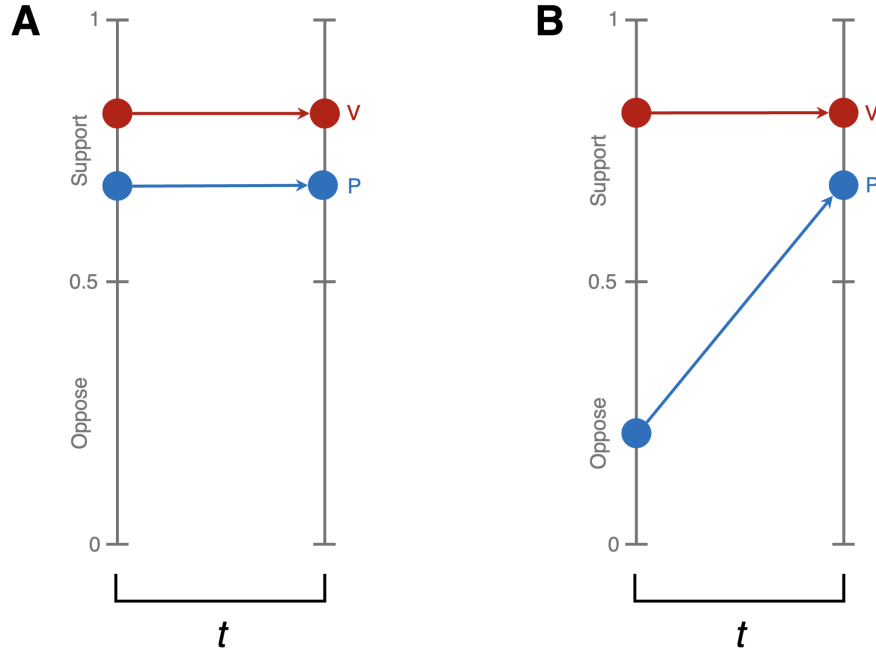
Country	Panelists	Policies	Waves	Period Lengths	Panel Length
Germany	26,683	48	23	$\bar{x} = 1.6$ years; $s = 1.39$	Oct. 2016 – Dec. 2021
Netherlands	1,670	15	11	$\bar{x} = 4.1$ years; $s = 2.62$	Dec. 2012 – Mar. 2024
United States	1,289	45	3	$\bar{x} = 2.6$ years; $s = 0.93$	Apr. 2008 – Aug. 2012

countries have high-quality panels with measures of basic values and many related policy opinions. Both the Dutch and U.S. panels use probability-based sampling designs, while the German panel draws its respondents from an online access panel using quotas for age, gender, and education. The countries also differ in potentially important ways: The United States’ two major parties may make the mapping from values to opinions relatively simple (Levendusky 2010), while the fragmented multi-party systems of Germany and the Netherlands may complicate that mapping (Brader, Tucker, and Duell 2013).

My empirical expectation is straightforward: If stability identifies meaningful opinions, responses consistent with a citizen’s values at one wave should change less by the next. Figure 1 illustrates this intuition for some period t , at whose beginning and end we observe a value and a related policy opinion: When the opinion (in blue) is highly consistent with the value (in red), it should change little over t (panel A); when it is less consistent, it should change more (panel B).

Concretely, consider two German panelists who both want “the state to be strong so it can defend its citizens”—one of the statements used to measure conservation values. When asked whether “Germany’s defence expenditure should be increased over the next few years,” the first agrees and the second disagrees. If the second panelist is just as likely as the first to hold their position at the next wave, then serving a citizen’s fundamental goals lends an opinion no staying power. In that case, stability cannot tell us which opinions reflect those goals and, thus, can steer government toward them.

Figure 1: Empirical Expectations



Note: This figure gives a stylized depiction of my empirical expectations. Red points represent a panelist's position on a value at the beginning and end of a period, t , while blue points represent their position on a related policy opinion.

To test this intuition, I restructure the panel data so each row captures panelist i 's opinion on policy p and its relationship to value v over period t , where a period joins any two waves (e.g., $t = w_1 \rightarrow w_2$ or $t = w_1 \rightarrow w_3$). I then compare the subsequent stability of opinions coded consistent and inconsistent at the period's start. Observations from panelists, values, policies, and periods are stacked, and fixed effects account for differences across cases.

Measurement

In each panel, I measure self-transcendence ($\alpha = 0.65$ – 0.78) and conservation ($\alpha = 0.59$ – 0.73) values by averaging items from a well-validated measure (Schwartz et al. 2001). Question wordings are in the Replication Materials.⁷ All three datasets measure each panelist's values

⁷The Replication Materials (online only) can be found here:

<https://drive.proton.me/urls/EJ1CJ9A1B4#cWU2wkCLTElx>

once at the panelist’s first wave (Germany), the panel’s beginning (the Netherlands), or the panel’s end (the United States). In line with past research, I assume that panelists’ basic values remain stable throughout the panel (Milfont, Milojev, and Sibley 2016; Vecchione et al. 2020). Appendix G validates this assumption with an additional panel dataset, and appendices H and C.3 confirm that value change does not bias my results.

For ease of interpretation, I measure value-consistency with a binary indicator equal to 1 if a citizen’s opinion falls on the value-consistent side of a Likert policy scale and 0 otherwise. For instance, consider this item from the Dutch panel:

“It should be made easier to obtain asylum in the Netherlands.” (Answers: Fully disagree, disagree, neither agree nor disagree, agree, and fully agree)

A citizen high in self-transcendence—i.e., someone with strong concern for others’ welfare—would be coded as consistent if they supported the policy and inconsistent if they opposed it or were ambivalent. I map self-transcendence and conservation to policy directions based on a systematic review of prior research, considering only relationships that manifest consistently across studies (see app. B). Appendix C.2 replicates my analyses using a purely empirical approach—regressing policy opinions on basic values to determine “what goes with what” (Converse 1964, 212)—with substantively similar results. To ensure comparability, I limit my main analyses to five- to seven-point Likert scales, though appendix C.1 replicates them with all items.

Importantly, my measure of value-consistency is behavioral: It records whether an opinion falls on the side of an issue that the literature ties to a value, not whether the citizen consciously reasons about the issue from that value. Under the representational definition above, that is the appropriate target, as citizens’ opinions are often shaped by their values without their being conscious of this influence (Jost 2021). It also matches how existing tests measure value-consistency—i.e., by comparing opinions to measured values (Feldman 2003; Goren et al. 2016), rather than asking citizens to report their reasons for supporting a policy. Even so, several robustness checks assess whether narrowing my sample to conscious value

reasoners would change my results. The consistency-stability gap remains statistically lower than, or indistinguishable from, a small effect among respondents for whom these values are most central (app. C.6) and for the issues whose cross-sectional value-policy correlations are strongest (app. C.7). The same holds among the highly educated, the politically interested, and the politically knowledgeable (app. C.8).

I code opinion stability two ways, again for ease of interpretation. The first measure equals 1 if a citizen gave the exact same response at both waves of a period and 0 otherwise. This simple measure of test-retest reliability is strict: Even a shift from “fully agree” to “agree” would register as instability, potentially attenuating true differences between value-consistent and -inconsistent opinions. The second measure is more lenient: It equals 1 if a citizen remained on the same side of the policy scale across both waves and 0 if they crossed to the other side or became ambivalent.

Statistical Models

To estimate the relationship between opinion stability and value-consistency, for each panel dataset, I run an OLS regression of the following form:⁸

$$OpinionStable_{i,p,t} = \beta_1 * ValueConsistent_{i,p,v,t} + \beta * Traits_{i,t} + \alpha_{p,v} + \delta_{s,p,t} + \epsilon$$

In this model, *OpinionStable* indicates whether panelist *i* maintained a stable opinion on policy issue *p* over period *t*. The primary independent variable, *ValueConsistent*, captures whether that opinion aligned with a specific value *v* at the start of the period. *Traits* is a matrix of subject-level controls: a panelist’s education, political interest, and age at the beginning of the period, and tendency to acquiesce to all value statements, regardless of their

⁸Logit models produce qualitatively similar results (see app. A).

content (Schwartz 1992).⁹ This last term minimizes measurement error in *ValueConsistent* and absorbs any tendency for the acquiescence-prone to hold less stable opinions.

To account for unobserved heterogeneity, the model incorporates two layers of fixed effects. First, “case” fixed effects (α) represent specific value-policy pairings, ensuring the model compares consistency and stability only within a given pairing. Second, “trend” fixed effects (δ) account for the initial stance (s) held on a policy issue (e.g., left, center, or right) during period t . This term allows for the possibility that certain positions are more unstable in general or during specific timeframes. For instance, moderate positions may be more unstable in general, and conservative stances on government intervention may have been more volatile following the 2008 financial crisis. These fixed effects also control for differences in the length of time between panelists’ responses.

This approach estimates a single coefficient (β_1) representing the general tendency for value-consistent opinions to be more stable over time. This estimand is correlational rather than causal, but prediction is sufficient to assess a proxy. To be sure, scholars care about stable opinions largely because they are presumed to be causally downstream of citizens’ values, and I have taken several steps to strengthen a causal reading: I control for other potential causes of opinion stability and, in the German and Dutch panels, alleviate concerns about reverse causality by ensuring my dependent variable is always measured after the independent variables. But my question is more pragmatic: Are the best available data consistent with a world in which stability and value-consistency reliably predict one another? If not, that alone should caution against treating stability as a proxy for meaningfulness.

⁹In the U.S. panel, political interest was asked at every wave, but only of the roughly two-thirds of panelists who received particular survey ballots. U.S. models that control for it therefore include only those panelists.

Stability Barely Tracks Value-Consistency

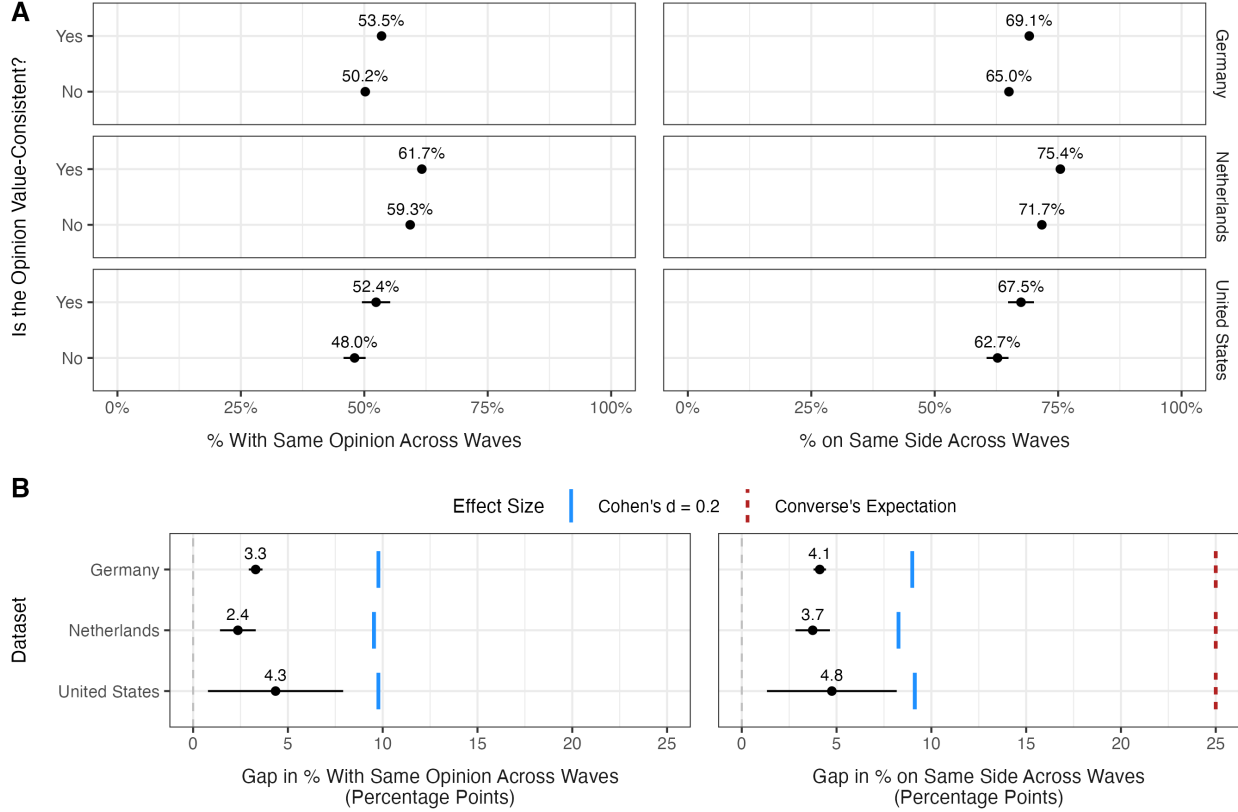
Panel A of Figure 2 plots the predicted stability of value-consistent and -inconsistent opinions—averaged across policy issues, values, and periods. It shows the predicted percentage of respondents who express the exact same opinion across waves (left) and the predicted percentage who remain on the same side of a policy scale across waves (right). Panel B plots the gap between these estimates against two benchmarks: a conventionally small effect (Cohen’s $d \leq 0.2$) and the gap Converse (1964) expected between citizens’ and legislators’ opinions. The latter is offered as historical context: Though comparing legislators with citizens is unlike comparing value-consistent and -inconsistent opinions within citizens, the expectation that launched this literature is a natural yardstick.

As seen on the left side of panel A in Figure 2, panelists in all three panels are only slightly more likely to hold on to value-consistent than -inconsistent opinions. For instance, in the U.S. panel, the model predicts that 48.0% (95% confidence interval [CI] 45.8–50.3) of panelists who voice a value-inconsistent opinion at one wave will voice the exact same opinion at a later one, compared to 52.4% (95% CI 49.5–55.2) of panelists who voice a value-consistent opinion. This amounts to an increase of roughly four percentage points ($p < 0.05$). In all three panels, the upper bound of the 95% confidence interval falls below a conventionally small effect size (Cohen’s $d \leq 0.2$).

The right side of panel A tells a similar story. In the U.S. panel, the model predicts that 62.7% (95% CI 60.5–64.9) of panelists with a value-inconsistent opinion will remain on the same side of a policy scale across waves, compared to 67.5% (95% CI 64.9–70.1) of those with a value-consistent opinion—a substantively small gap of 4.8 percentage points ($p < 0.01$). The gap is even smaller in the German and Dutch panels, at 4.1 and 3.7 percentage points, respectively ($p < 0.001$ in both cases). Again, every upper bound falls below the small-effect threshold.

These gaps are at most one-fifth of the 25-percentage-point difference Converse (1964) expected between ordinary citizens (roughly 65% of whom stayed on the same side of a

Figure 2: Value-Consistent Opinions Are Only Slightly More Stable Than Value-Inconsistent Ones

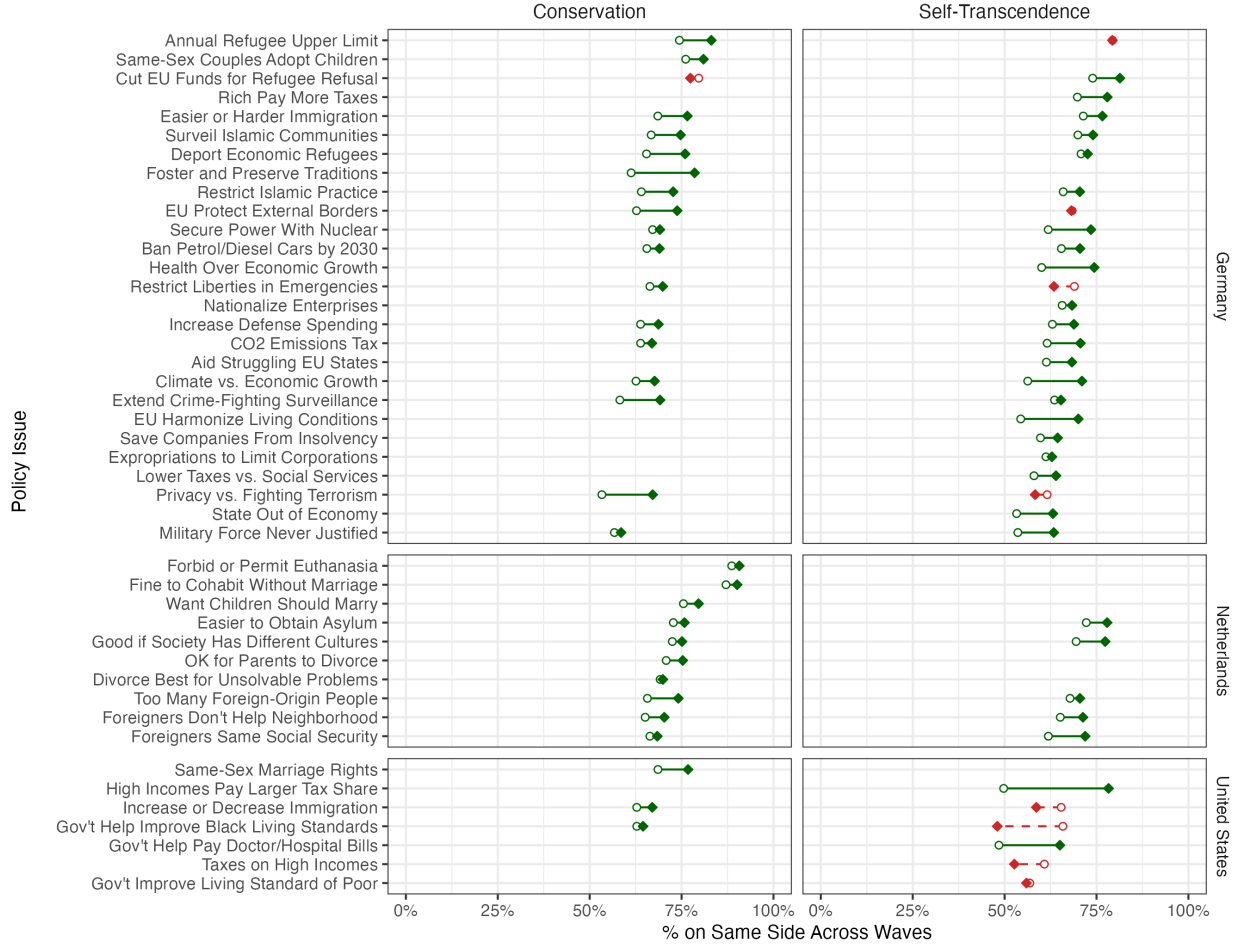


Note: Panel A shows the predicted stability of value-consistent and -inconsistent opinions. Panel B shows the gap between these estimates alongside Cohen's $d = 0.2$ (solid blue) and Converse's expected citizen-legislator gap (dashed red). Data are weighted to give each panelist and, as nearly as the data allow, each value-policy pair equal weight. Lines represent 95% confidence intervals from wild-bootstrap standard errors clustered by panelist.

policy issue over two years) and legislators (about 90%). Importantly, the average period lengths in the panel datasets straddle Converse's two-year benchmark (Germany: $\bar{x} = 1.6$ years; Netherlands: $\bar{x} = 4.1$ years; United States: $\bar{x} = 2.6$ years). In any case, while longer intervals between waves reduce average opinion stability, they barely widen the gap between value-consistent and -inconsistent opinions (see app. E).

But does even this small gap make stability a useful proxy for meaningfulness? A useful proxy should substantially update one's belief about the underlying quantity. Formally, let C denote that an opinion is value-consistent and S denote that it is stable. By Bayes's theorem, $P(C|S) = \frac{P(S|C) \times P(C)}{P(S)}$. Using the predicted stability rates in panel A of Figure 2 and the

Figure 3: The Stability Gap Is Small in Most Cases and Reverses in Some



Note: This figure shows the predicted stability of value-consistent (filled diamonds) and -inconsistent (open circles) opinions for each case, broken out by value and dataset. Solid green lines indicate cases where value-consistent opinions are more stable, while dashed red lines indicate cases where the relationship reverses. Data are weighted to give each panelist equal weight.

empirical base rate of value-consistent opinions in each sample, I subtract $P(C|\neg S)$ from this quantity and find that stability shifts the probability that an opinion is value-consistent by no more than five percentage points.¹⁰ In other words, whether an opinion is stable tells us little about whether it aligns with a citizen's values.

¹⁰For same-side stability, $P(C|S) - P(C|\neg S)$ equals 0.044, 0.046, and 0.050 in Germany, the Netherlands, and the United States, respectively. For same-opinion stability, it equals 0.032, 0.024, and 0.042.

These averages, however, obscure notable differences across value-policy pairings. Figure 3 plots the predicted stability of value-consistent (filled diamonds) and -inconsistent (open circles) opinions for each case, broken out by value and dataset (see app. D for model details).

Two patterns stand out. First, while value-consistent opinions are more stable in most cases, the gap is typically small in the German and Dutch panels: The filled diamonds sit only slightly to the right of the open circles. Second, in a few cases—across both value types—value-consistent opinions are less stable than value-inconsistent ones.

The U.S. case-level estimates are more heterogeneous and imprecise: Larger positive and negative estimates partly offset one another rather than producing a uniformly small pattern, likely because the U.S. panel has fewer panelists and waves. In the German and Dutch panels, by contrast, estimates are more precise, and the average effect is small because the case-level effects are themselves small. Either way, opinion stability is a poor proxy for value-consistency: The relationship is too weak to be useful in the German and Dutch panels and too noisy to be reliable in the U.S. panel.

Supplementary Analyses

Thus far, the motivating assumption behind research on opinion stability has poor empirical support: Value-consistent opinions are only slightly more stable than value-inconsistent ones. Yet small correlations may have various causes, such as measurement error. In the online appendix, I run several supplementary analyses to assess the robustness of my findings to three concerns. I summarize these checks in Table 4 and briefly describe each below. None substantively alters the findings.

Does Error Attenuate the Consistency-Stability Link?

A natural question is whether error in my value measurements attenuates the association between value-consistency and opinion stability. Recall that each panel measures values once,

Table 4: Supplementary Analyses

Concern	Analysis
Measurement error may attenuate the consistency-stability link.	Compare stability of basic values and policy opinions. Test whether cross-sectional value-policy correlations shift over time. Subset to participants whose values are least likely to change. Construct indexes of policy opinions (Netherlands and United States). Measure value-consistency and opinion stability as continuous variables.
The selected value may be less relevant in some observations.	Subset to participants who most strongly endorse or reject self-transcendence (conservation). Subset to cases with the strongest cross-sectional value-policy correlations.
Stability may discriminate better in more sophisticated groups.	Subset to highly educated participants. Subset to politically interested participants. Subset to politically knowledgeable participants (Germany).

which requires me to impute citizens' values at other waves. If values and policy opinions tend to drift in the same direction over time, this approach could weaken the estimated consistency-stability link.

Several lines of evidence suggest it does not. Prior work shows that basic values remain stable for long periods, which suggests any measurement error is small (Milfont, Milojev, and Sibley 2016; Vecchione et al. 2016, 2020). Using an additional panel with repeated value measures, I confirm that basic values fluctuate little—and far less than most policy opinions (app. G). Appendix H tests a further implication: If values shifted, the average cross-sectional association between values and policy opinions should weaken as the value measurements grow stale. Little decay appears. Finally, results are similar when the analysis is limited to middle-aged panelists (app. C.3), who are least likely to exhibit value change (Milfont, Milojev, and Sibley 2016).

Three further checks address error in the opinion measures. First, in the Dutch and U.S. panels, combining related policy opinions into indexes, which averages away item-level noise (Ansolabehere, Rodden, and Snyder 2008), leaves the stability gap small (app. C.4).

Second, replacing my binary codings with continuous measures of value-consistency and opinion stability yields similarly modest effects (app. C.5). Third, extending the analysis to all available policy items—including the shorter scales excluded from the main specification—produces comparable findings (app. C.1).

Is the Selected Value Relevant?

Although self-transcendence and conservation values are known to structure policy opinions (see app. B), the consistency-stability link could be diluted if the selected value is irrelevant in some observations—e.g., because the respondent cares little about it. To test this, I calculate a standard measure of value importance: the distance between a respondent’s endorsement of a given value and their average endorsement of all measured values (Schwartz et al. 2001).¹¹ For each value-policy pairing, I then subset to respondents in the top third of value importance and re-estimate the relevant models. The results, shown in appendix C.6, again indicate that the consistency-stability link is weak.

It is also possible that basic values are too removed from politics for the consistency-stability link to emerge. Restricting the analysis to value-policy pairs with the strongest cross-sectional associations (the top third in each panel) does not substantively alter the findings (app. C.7).

Does Stability Discriminate Better Among the Sophisticated?

Finally, my measure of value-consistency may not isolate opinions formed through conscious, value-based reasoning. For some citizens, the alignment may be purely incidental. Given that

¹¹That is, I calculate each participant’s average level of agreement with all basic values, not just self-transcendence and conservation values. Then, I subtract this cross-value average from the panelist’s ratings of self-transcendence and conservation values, respectively, and take the absolute value.

policy knowledge tends to be low (e.g., Gilens 2001), many citizens may not recognize which opinions are consistent with their values (Zaller 1992), and stability may only discriminate value-consistent from -inconsistent opinions among citizens who are equipped to connect values to concrete policies (Delli Carpini and Keeter 1996). To test this, appendix C.8 replicates my analyses among the most educated and politically interested panelists and, in the German panel, the most politically knowledgeable. The stability gap between value-consistent and -inconsistent opinions is similar in these subgroups, and it remains small in substantive terms. This suggests that even among more politically sophisticated citizens, opinion stability is a poor proxy for value-consistency.

Discussion and Conclusion

This article has tested an inference that animates some of the most thoughtful and empirically rigorous work in political science: that an opinion’s over-time stability indicates whether it is “meaningful,” or capable of guiding public policy toward citizens’ goals. Instability troubles scholars because it seems to leave elected officials without meaningful preferences to represent. In particular, an opinion cannot steer government toward what a citizen wants unless it is consistent with their most basic goals, or “values.”

The test returns a clear answer. Across three countries with starkly different party systems, value-consistent opinions are only modestly more stable than value-inconsistent ones, and learning that an opinion is stable changes the probability that it is value-consistent by no more than five percentage points. These results hold across a battery of robustness checks.

How can this be, when values are consistently associated with policy opinions in cross-sectional data (see app. B)? The answer, I suggest, is that the assumed link holds only if two conditions jointly obtain: The mapping from values to policy positions must stay fixed, and no force besides values may anchor the opinion. Prior work gives strong reasons to doubt each.

First, value-policy mappings—i.e., which position on a policy issue serves a particular value—can shift, so staying value-consistent can require changing one’s opinion. Return to the two German panelists above. Citizens higher in conservation values do express greater support for defense spending, and studies find this cross-sectional association across countries and contexts (e.g., Rathbun et al. 2016; Schwartz et al. 2014). But it does not follow that the first panelist’s support should remain fixed. Citizens’ policy opinions react thermostatically to existing policy (Wlezien 1995): If Germany had recently raised its defense budget substantially, further increases might seem unnecessary even to those who prioritize security. Opinion change, in other words, can reflect principled thinking rather than its absence.

Mappings can also shift because policies engage multiple values, and frames change which connection is salient (Nelson, Clawson, and Oxley 1997). When a policy conflict pits two values against one another, citizens side with the value they prioritize (Tetlock 1986). Yet if political discourse highlights only one value, citizens can only ground their opinions in that value. As such, a citizen who supports a policy protecting civil liberties when both freedom and security are salient could later oppose the same policy when only security is salient, without changing either value. Media coverage can alter which values citizens bring to bear on policy issues (Brewer 2002), and the framing environment changes over time (Chong and Druckman 2007). And because citizens encounter different information, circumstances, and frames, such remapping need not be uniform: Value-driven opinion changes may offset one another in the aggregate, leaving little trace in the cross-sectional association between values and policy opinions.

A second potential explanation is rival anchors. Values are not the only force that can hold policy opinions in place (for a review, see Mintz, Valentino, and Wayne 2021). Partisan attachments, in particular, may be a powerful source of opinion stability (Elder and O’Brian 2022; Freeder, Lenz, and Turney 2019). Citizens often feel emotionally attached to their political parties (Huddy, Mason, and Aarøe 2015), and an abundance of experiments shows that they readily adopt or defend policy opinions, regardless of ideological content, when their

party endorses them (for a review, see Bullock 2020). Thus, under typical circumstances—e.g., when citizens agree with their parties (Dalton 2017), parties’ issue positions remain stable (Dalton 2016), and citizens’ party attachments persist (Green and Platzman 2024)—partisan loyalty could stabilize opinions over time, whether or not those opinions reflect citizens’ values. Indeed, as Druckman, Fein, and Leeper (2012) write, “[W]hen opinions display stability, it may reflect an underlying bias rather than considered opinions” (445).

Together, the two routes could compress the stability gap from both sides: A moving mapping can reduce stability among initially consistent opinions, while rival anchors can increase stability among initially inconsistent opinions. Which route operates where, however, is a question my panel data are ill-suited to answer. The data do not measure how citizens perceive value-policy alignments, so I cannot estimate what share of opinion changes follow shifts in perceived mappings. And because party-loyal and value-consistent opinions are often observationally equivalent—and citizens may select parties on the basis of their values—modeling both simultaneously risks post-treatment bias that understates the effects of values. Future work should measure perceived mappings repeatedly. Alternatively, one could experimentally manipulate the value-consistency and party-loyalty of competing positions on a non-salient issue and observe how each affects stability.

That said, other limitations of my data are noteworthy. For one, the study observes consistency with one selected value at a time—not a person’s full value system—so a response coded as inconsistent may in fact advance an unmeasured, competing value. That said, when issues implicate both self-transcendence and conservation values, modeling both simultaneously leaves the consistency-stability gap small (app. C.9). Relatedly, my measure of value-consistency is behavioral, requiring no conscious value reasoning. But where such reasoning is most plausible, the consistency-stability gap remains statistically lower than, or indistinguishable from, a small effect (app. C.6). Finally, each panel measures basic values only once, requiring me to impute values across waves. However, several additional analyses suggest that this choice does not bias my results (apps. C.3, G, and H).

These limitations bound my estimates, but they do not restore opinion stability as a proxy for value-consistency. If we wish to evaluate whether citizens hold value-consistent opinions—and we should, if we care about democratic representation—we need better tools. A more promising path than asking how often citizens’ opinions change is to ask whether citizens have the skills to form value-consistent opinions (Druckman 2014)—e.g., whether they respond to specific policy characteristics in ways that reflect their values. This skills-based approach does not require stability to proxy for value-consistency or researchers to anticipate every relevant value.

For six decades, the instability of citizens’ opinions has been read as evidence that citizens are incompetent and that elected officials lack preferences to represent. That reading asked stability to certify something it cannot detect. Discarding stability as a diagnostic does not establish that citizens are competent; it only removes one central piece of evidence that they are not. Nonetheless, we have been judging citizens by a standard that cannot support our judgment. Our judgment should wait until we adopt one that can.

References

- Achen, Christopher H. 1975. "Mass Political Attitudes and the Survey Response." *American Political Science Review* 69 (4): 1218–1231.
- Achen, Christopher H., and Larry M. Bartels. 2017. *Democracy for Realists: Why Elections Do Not Produce Responsive Government*. Princeton, NJ: Princeton University Press.
- Ansolabehere, Stephen, and Shiro Kuriwaki. 2022. "Congressional Representation: Accountability from the Constituent's Perspective." *American Journal of Political Science* 66 (1): 123–139.
- Ansolabehere, Stephen, Jonathan Rodden, and James M. Snyder. 2008. "The Strength of Issues: Using Multiple Measures to Gauge Preference Stability, Ideological Constraint, and Issue Voting." *American Political Science Review* 102 (2): 215–232.
- Arceneaux, Kevin, Bert N. Bakker, Neil Fasching, and Yphtach Lelkes. 2025. "A Critical Evaluation and Research Agenda for the Study of Psychological Dispositions and Political Attitudes." *Political Psychology* 46 (S1): 7–37.
- Brader, Ted, Joshua A. Tucker, and Dominik Duell. 2013. "Which Parties Can Lead Opinion? Experimental Evidence on Partisan Cue Taking in Multiparty Democracies." *Comparative Political Studies* 46 (11): 1485–1517.
- Brewer, Paul R. 2002. "Framing, Value Words, and Citizens' Explanations of Their Issue Opinions." *Political Communication* 19 (3): 303–316.
- Bullock, John G. 2020. "Party Cues." In Elizabeth Suhay, Bernard Grofman, and Alexander H. Trechsel, eds., *The Oxford Handbook of Electoral Persuasion*. New York: Oxford University Press, 128–150.
- Butler, Daniel M., and David W. Nickerson. 2011. "Can Learning Constituency Opinion Affect How Legislators Vote? Results from a Field Experiment." *Quarterly Journal of Political Science* 6 (1): 55–83.
- Chong, Dennis, and James N. Druckman. 2007. "Framing Public Opinion in Competitive Democracies." *American Political Science Review* 101 (4): 637–655.

- Collins, Patricia R., Julie A. Lee, Joanne N. Sneddon, and Anna K. Döring. 2017. "Examining the Consistency and Coherence of Values in Young Children Using a New Animated Values Instrument." *Personality and Individual Differences* 104:279–285.
- Conner, Mark, Sarah Wilding, and Paul Norman. 2022. "Testing Predictors of Attitude Strength as Determinants of Attitude Stability and Attitude-Behaviour Relationships: A Multi-behaviour Study." *European Journal of Social Psychology* 52 (4): 656–668.
- Converse, Philip E. 1964. "The Nature of Belief Systems in Mass Publics." In David E. Apter, ed., *Ideology and Discontent*. New York: Free Press, 206–261.
- Converse, Philip E. 2000. "Assessing the Capacity of Mass Electorates." *Annual Review of Political Science* 3 (1): 331–353.
- Dahl, Robert A. 1971. *Polyarchy: Participation and Opposition*. New Haven, CT: Yale University Press.
- Dalton, Russell J. 2016. "Stability and Change in Party Issue Positions: The 2009 and 2014 European Elections." *Electoral Studies* 44:525–534.
- Dalton, Russell J. 2017. "Party Representation across Multiple Issue Dimensions." *Party Politics* 23 (6): 609–622.
- Delli Carpini, Michael X., and Scott Keeter. 1996. *What Americans Know about Politics and Why It Matters*. New Haven, CT: Yale University Press.
- Döring, Anna K., Shalom H. Schwartz, Jan Cieciuch, Patrick J. F. Groenen, Valentina Glatzel, Justyna Harasimczuk, Nicole Janowicz, Maya Nyagolova, E. Rebecca Scheefer, Matthias Allritz, Taciano L. Milfont, and Wolfgang Bilsky. 2015. "Cross-Cultural Evidence of Value Structures and Priorities in Childhood." *British Journal of Psychology* 106 (4): 675–699.
- Druckman, James N. 2001. "The Implications of Framing Effects for Citizen Competence." *Political Behavior* 23 (3): 225–256.
- Druckman, James N. 2014. "Pathologies of Studying Public Opinion, Political Communication, and Democratic Responsiveness." *Political Communication* 31 (3): 467–492.

- Druckman, James N., Jordan Fein, and Thomas J. Leeper. 2012. "A Source of Bias in Public Opinion Stability." *American Political Science Review* 106 (2): 430–454.
- Elder, Elizabeth Mitchell, and Neil A. O'Brian. 2022. "Social Groups as the Source of Political Belief Systems: Fresh Evidence on an Old Theory." *American Political Science Review* 116 (4): 1407–1424.
- Ellis, Christopher, and James A. Stimson. 2012. *Ideology in America*. Cambridge, England: Cambridge University Press.
- Enke, Benjamin, Ricardo Rodríguez-Padilla, and Florian Zimmermann. 2023. "Moral Universalism and the Structure of Ideology." *Review of Economic Studies* 90 (4): 1934–1962.
- Erikson, Robert S. 1979. "The SRC Panel Data and Mass Political Attitudes." *British Journal of Political Science* 9 (1): 89–114.
- Feldman, Stanley. 1988. "Structure and Consistency in Public Opinion: The Role of Core Beliefs and Values." *American Journal of Political Science* 32 (2): 416–440.
- Feldman, Stanley. 1989. "Measuring Issue Preferences: The Problem of Response Instability." *Political Analysis* 1:25–60.
- Feldman, Stanley. 2003. "Values, Ideology, and the Structure of Political Attitudes." In David O. Sears, Leonie Huddy, and Robert Jervis, eds., *Oxford Handbook of Political Psychology*. New York: Oxford University Press, 477–508.
- Freder, Sean, Gabriel S. Lenz, and Shad Turney. 2019. "The Importance of Knowing 'What Goes with What': Reinterpreting the Evidence on Policy Attitude Stability." *Journal of Politics* 81 (1): 274–290.
- Gilens, Martin. 2001. "Political Ignorance and Collective Policy Preferences." *American Political Science Review* 95 (2): 379–396.
- GLÉS. 2023. "GLÉS Panel 2016–2021, Waves 1–21 (ZA6838; Version 6.0.0) [Dataset]." Cologne: GESIS – Leibniz Institute for the Social Sciences. <https://doi.org/10.4232/1.14114> (accessed October 5, 2026).

- Goren, Paul. 2005. "Party Identification and Core Political Values." *American Journal of Political Science* 49 (4): 881–896.
- Goren, Paul. 2012. *On Voter Competence*. New York: Oxford University Press.
- Goren, Paul, Christopher M. Federico, and Miki Caul Kittilson. 2009. "Source Cues, Partisan Identities, and Political Value Expression." *American Journal of Political Science* 53 (4): 805–820.
- Goren, Paul, Harald Schoen, Jason Reifler, Thomas Scotto, and William Chittick. 2016. "A Unified Theory of Value-Based Reasoning and U.S. Public Opinion." *Political Behavior* 38 (4): 977–997.
- Green, Donald P., and Paul Platzman. 2024. "Partisan Stability during Turbulent Times: Evidence from Three American Panel Surveys." *Political Behavior* 46 (1): 377–403.
- Horowitz, Michael C., and Matthew S. Levendusky. 2011. "Drafting Support for War: Conscription and Mass Support for Warfare." *Journal of Politics* 73 (2): 524–534.
- Howe, Lauren C., and Jon A. Krosnick. 2017. "Attitude Strength." *Annual Review of Psychology* 68 (1): 327–351.
- Huddy, Leonie, Lilliana Mason, and Lene Aarøe. 2015. "Expressive Partisanship: Campaign Involvement, Political Emotion, and Partisan Identity." *American Political Science Review* 109 (1): 1–17.
- Inglehart, Ronald. 1985. "Aggregate Stability and Individual-Level Flux in Mass Belief Systems: The Level of Analysis Paradox." *American Political Science Review* 79 (1): 97–116.
- Jost, John T. 2021. *Left and Right: The Psychological Significance of a Political Distinction*. New York: Oxford University Press.
- Kinder, Donald R., and Nathan P. Kalmoe. 2017. *Neither Liberal nor Conservative: Ideological Innocence in the American Public*. Chicago, IL: University of Chicago Press.
- Krosnick, Jon A. 1988. "Attitude Importance and Attitude Change." *Journal of Experimental Social Psychology* 24 (3): 240–255.

- Krosnick, Jon A. 1990. "Government Policy and Citizen Passion: A Study of Issue Publics in Contemporary America." *Political Behavior* 12 (1): 59–92.
- Lau, Richard R., and David P. Redlawsk. 1997. "Voting Correctly." *American Political Science Review* 91 (3): 585–598.
- Lauderdale, Benjamin E., Chris Hanretty, and Nick Vivyan. 2018. "Decomposing Public Opinion Variation into Ideology, Idiosyncrasy, and Instability." *Journal of Politics* 80 (2): 707–712.
- Leeper, Thomas J. 2014. "Are Important Attitudes More Stable? No, Not Really." Unpublished manuscript.
- Levendusky, Matthew S. 2010. "Clearer Cues, More Consistent Voters: A Benefit of Elite Polarization." *Political Behavior* 32 (1): 111–131.
- Luttrell, Andrew, and LaCount J. Togans. 2021. "The Stability of Moralized Attitudes over Time." *Personality and Social Psychology Bulletin* 47 (4): 551–564.
- Malka, Ariel, and Yphtach Lelkes. 2010. "More Than Ideology: Conservative–Liberal Identity and Receptivity to Political Cues." *Social Justice Research* 23 (2–3): 156–188.
- McCann, James A. 1997. "Electoral Choices and Core Value Change: The 1992 Presidential Campaign." *American Journal of Political Science* 41 (2): 564–583.
- Milfont, Taciano L., Petar Milojev, and Chris G. Sibley. 2016. "Values Stability and Change in Adulthood: A 3-Year Longitudinal Study of Rank-Order Stability and Mean-Level Differences." *Personality and Social Psychology Bulletin* 42 (5): 572–588.
- Mintz, Alex, Nicholas A. Valentino, and Carly Wayne. 2021. *Beyond Rationality: Behavioral Political Science in the 21st Century*. Cambridge, England: Cambridge University Press.
- Mummolo, Jonathan, Erik Peterson, and Sean Westwood. 2021. "The Limits of Partisan Loyalty." *Political Behavior* 43 (3): 949–972.
- Nelson, Thomas E., Rosalee A. Clawson, and Zoe M. Oxley. 1997. "Media Framing of a Civil Liberties Conflict and Its Effect on Tolerance." *American Political Science Review* 91 (3): 567–583.

- Page, Benjamin I., and Robert Y. Shapiro. 1992. *The Rational Public: Fifty Years of Trends in Americans' Policy Preferences*. Chicago, IL: University of Chicago Press.
- Peffley, Mark A., and Jon Hurwitz. 1985. "A Hierarchical Model of Attitude Constraint." *American Journal of Political Science* 29 (4): 871–890.
- Peffley, Mark, and Jon Hurwitz. 1993. "Models of Attitude Constraint in Foreign Affairs." *Political Behavior* 15 (1): 61–90.
- Price, Vincent, and Peter Neijens. 1997. "Opinion Quality in Public Opinion Research." *International Journal of Public Opinion Research* 9 (4): 336–360.
- Pronin, Emily. 2009. "The Introspection Illusion." In Mark P. Zanna, ed., *Advances in Experimental Social Psychology*. Vol. 41. Amsterdam: Academic Press, 1–67.
- Rathbun, Brian C., Joshua D. Kertzer, Jason Reifler, Paul Goren, and Thomas J. Scotto. 2016. "Taking Foreign Policy Personally: Personal Values and Foreign Policy Attitudes." *International Studies Quarterly* 60 (1): 124–137.
- Rokeach, Milton. 1973. *The Nature of Human Values*. New York: Free Press.
- Sabl, Andrew. 2015. "The Two Cultures of Democratic Theory: Responsiveness, Democratic Quality, and the Empirical-Normative Divide." *Perspectives on Politics* 13 (2): 345–365.
- Scherpenzeel, Annette C., and Marcel Das. 2010. "'True' Longitudinal and Probability-Based Internet Panels: Evidence from the Netherlands." In Marcel Das, Peter Ester, and Lars Kaczmirek, eds., *Social and Behavioral Research and the Internet: Advances in Applied Methods and Research Strategies*. Boca Raton, FL: Taylor & Francis, 77–104.
- Schwartz, Shalom H. 1992. "Universals in the Content and Structure of Values: Theoretical Advances and Empirical Tests in 20 Countries." In Mark P. Zanna, ed., *Advances in Experimental Social Psychology*. Vol. 25. San Diego, CA: Academic Press, 1–65.
- Schwartz, Shalom H. 2012. "An Overview of the Schwartz Theory of Basic Values." *Online Readings in Psychology and Culture* 2 (1). <https://doi.org/10.9707/2307-0919.1116>.
- Schwartz, Shalom H., Gian Vittorio Caprara, Michele Vecchione, Paul Bain, Gabriel Bianchi, Maria Giovanna Caprara, Jan Cieciuch, Hasan Kirmanoglu, Cem Baslevent, Jan-Erik

- Lönnqvist, Catalin Mamali, Jorge Manzi, Vassilis Pavlopoulos, Tetyana Posnova, Harald Schoen, Jo Silvester, Carmen Tabernero, Claudio Torres, Markku Verkasalo, Eva Vondráková, Christian Welzel, and Zbigniew Zaleski. 2014. “Basic Personal Values Underlie and Give Coherence to Political Values: A Cross National Study in 15 Countries.” *Political Behavior* 36 (4): 899–930.
- Schwartz, Shalom H., Jan Cieciuch, Michele Vecchione, Eldad Davidov, Ronald Fischer, Constanze Beierlein, Alice Ramos, Markku Verkasalo, Jan-Erik Lönnqvist, Kursad Demirutku, Ozlem Dirilen-Gumus, and Mark Konty. 2012. “Refining the Theory of Basic Individual Values.” *Journal of Personality and Social Psychology* 103 (4): 663–688.
- Schwartz, Shalom H., Gila Melech, Arielle Lehmann, Steven Burgess, Mari Harris, and Vicki Owens. 2001. “Extending the Cross-Cultural Validity of the Theory of Basic Human Values with a Different Method of Measurement.” *Journal of Cross-Cultural Psychology* 32 (5): 519–542.
- Sears, David O., and Carolyn L. Funk. 1991. “The Role of Self-Interest in Social and Political Attitudes.” In Mark P. Zanna, ed., *Advances in Experimental Social Psychology*. Vol. 24. San Diego, CA: Academic Press, 1–91.
- Smith, Tom W., Peter V. Marsden, and Michael Hout. 2018. “General Social Survey 2008–2012 Panel, Release 6 [Dataset].” Chicago: NORC at the University of Chicago. <https://gss.norc.org/us/en/gss/get-the-data.html> (accessed October 5, 2026).
- Sniderman, Paul M., and John Bullock. 2004. “A Consistency Theory of Public Opinion and Political Choice: The Hypothesis of Menu Dependence.” In Willem E. Saris and Paul M. Sniderman, eds., *Studies in Public Opinion: Attitudes, Nonattitudes, Measurement Error, and Change*. Princeton, NJ: Princeton University Press, 337–357.
- Sniderman, Paul M., and Sean M. Theriault. 2004. “The Structure of Political Argument and the Logic of Issue Framing.” In Willem E. Saris and Paul M. Sniderman, eds., *Studies in Public Opinion: Attitudes, Nonattitudes, Measurement Error, and Change*. Princeton, NJ: Princeton University Press, 133–165.

- Taber, Charles S., and Milton Lodge. 2006. "Motivated Skepticism in the Evaluation of Political Beliefs." *American Journal of Political Science* 50 (3): 755–769.
- Tetlock, Philip E. 1986. "A Value Pluralism Model of Ideological Reasoning." *Journal of Personality and Social Psychology* 50 (4): 819–827.
- Turner-Zwinkels, Felicity M., and Mark J. Brandt. 2025. "Ideology Strength versus Party Identity Strength: Ideology Strength Is the Key Predictor of Attitude Stability." *Personality and Social Psychology Bulletin* 51 (1): 125–138.
- Vecchione, Michele, Gianvittorio Caprara, Francesco Dentale, and Shalom H. Schwartz. 2013. "Voting and Values: Reciprocal Effects over Time." *Political Psychology* 34 (4): 465–485.
- Vecchione, Michele, Shalom Schwartz, Guido Alessandri, Anna K. Döring, Valeria Castellani, and Maria Giovanna Caprara. 2016. "Stability and Change of Basic Personal Values in Early Adulthood: An 8-Year Longitudinal Study." *Journal of Research in Personality* 63:111–122.
- Vecchione, Michele, Shalom H. Schwartz, Eldad Davidov, Jan Cieciuch, Guido Alessandri, and Gilda Marsicano. 2020. "Stability and Change of Basic Personal Values in Early Adolescence: A 2-Year Longitudinal Study." *Journal of Personality* 88 (3): 447–463.
- Vivyan, Nick, Benjamin E. Lauderdale, and Chris Hanretty. 2026. *Idiosyncratic Issue Opinion and Political Choice*. Oxford: Oxford University Press.
- Wlezien, Christopher. 1995. "The Public as Thermostat: Dynamics of Preferences for Spending." *American Journal of Political Science* 39 (4): 981–1000.
- Zaller, John R. 1992. *The Nature and Origins of Mass Opinion*. Cambridge, England: Cambridge University Press.

Online Appendix for “Don’t Judge Citizens for Changing Their Opinions”

Appendix A: Tabular Models	1
Appendix B: Identifying Value-Policy Pairs	3
Appendix C: Robustness Checks	6
C.1 Results When Including Shorter Scales	6
C.2 Results With Empirical Alignment	6
C.3 Results Among the Middle-Aged	8
C.4 Results Using Indexes of Policy Opinions	8
C.5 Results When Using Continuous Measures	9
C.6 Results When Values Are Most Important	9
C.7 Results for Strongest Value-Policy Cases	10
C.8 Results Among Sophisticated Panelists	10
C.9 Results When Issues Implicate Both Values	11
Appendix D: Modeling Case-Wise Heterogeneity	12
Appendix E: Does Time Affect the Stability Gap?	12
Appendix F: Stability of Political Values	14
Appendix G: Stability of Basic Values	16
Appendix H: Value-Policy Correlations Over Time	18

Appendix A: Tabular Models

Table A1 reports the full OLS models behind the main results, and Table A2 reports their logit counterparts.

Table A1: OLS Models of Opinion Stability

	Germany		Netherlands		United States	
	Same Opinion	Same Side	Same Opinion	Same Side	Same Opinion	Same Side
Value-Consistent	0.033 (0.002)	0.041 (0.002)	0.024 (0.005)	0.037 (0.005)	0.043 (0.018)	0.048 (0.017)
Education	0.023 (0.004)	0.040 (0.004)	0.056 (0.014)	0.048 (0.012)	0.052 (0.030)	0.077 (0.030)
Political Interest	0.020 (0.005)	0.031 (0.005)	0.000 (0.013)	0.029 (0.011)	0.019 (0.025)	0.020 (0.023)
Age	0.000 (0.000)	0.001 (0.000)	0.001 (0.000)	0.000 (0.000)	0.001 (0.001)	0.001 (0.001)
Acquiescence	-0.070 (0.012)	-0.054 (0.011)	-0.101 (0.030)	-0.040 (0.025)	-0.061 (0.074)	-0.084 (0.072)
Observations	10,190,569	10,190,569	768,551	768,551	11,084	11,084
RMSE	0.49	0.45	0.48	0.41	0.49	0.46
Cohen's $d = 0.2$	0.098	0.090	0.095	0.083	0.098	0.091

Note: Each model includes case and trend fixed effects. Data weighted to give each panelist and, as nearly as the data allow, each value-policy pair equal weight. Wild-bootstrap standard errors clustered by panelist in parentheses. The Cohen row gives a small-effect reference on the same 0–1 outcome scale as the coefficients.

Table A2: Logit Models of Opinion Stability

	Germany		Netherlands		United States	
	Same Opinion	Same Side	Same Opinion	Same Side	Same Opinion	Same Side
Value-Consistent	0.141 (0.008)	0.218 (0.009)	0.109 (0.022)	0.264 (0.031)	0.190 (0.080)	0.243 (0.092)
Education	0.096 (0.018)	0.199 (0.020)	0.245 (0.061)	0.279 (0.068)	0.223 (0.127)	0.373 (0.145)
Political Interest	0.083 (0.021)	0.152 (0.023)	0.001 (0.057)	0.171 (0.065)	0.079 (0.105)	0.097 (0.111)
Age	0.002 (0.000)	0.004 (0.000)	0.003 (0.001)	−0.002 (0.001)	0.004 (0.002)	0.004 (0.003)
Acquiescence	−0.294 (0.046)	−0.265 (0.051)	−0.449 (0.129)	−0.282 (0.141)	−0.262 (0.313)	−0.399 (0.347)
Observations	10,190,569	10,190,569	768,551	768,551	11,081	11,081
RMSE	0.49	0.45	0.48	0.41	0.49	0.46

Note: Each model includes case and trend fixed effects. Data weighted to give each panelist and, as nearly as the data allow, each value-policy pair equal weight. Standard errors clustered by panelist in parentheses.

Appendix B: Identifying Value-Policy Pairs

Table B1 maps policy areas to the values and lists the supporting studies. The search and coding protocol follows the table.

Table B1: “Social” Values and Related Policy Opinions

Policy Area	S.T.	CON.	Sources
Welfare, redistribution, and free enterprise (e.g., higher taxes on the rich)	Left	Mixed	Abdallah (2023); Arikan and Sekercioglu (2019); Reinl et al. (2024); Schwartz, Caprara, and Vecchione (2010); Schwartz et al. (2014); Wu and Molden (2024)
Environmental protection (e.g., CO ₂ taxes)	Left	Right	Abdallah (2023); Bergquist et al. (2022); Davidov and Meuleman (2012); Grunert and Juhl (1995); Milfont and Gouveia (2006); Nilsson, von Borgstede, and Biel (2004); Oh, Hudson, and Hudson (2025); Perlaviciute and Steg (2015); Schwartz (2012); Smith and Hempel (2022); Steg, Dreijerink, and Abrahamse (2005); Steg et al. (2011); Stern et al. (1995); Stern, Dietz, and Guagnano (1998)
Immigration	Left	Right	Abdallah (2023); Ahmadinia (2025); Cichocki and Jabkowski (2019); Davidov et al. (2008); Dennison, Davidov, and Seddig (2020); Messing and Ságvári (2021); Miglietta, Tartaglia, and Loera (2018); Schwartz (2007a); Schwartz (2007b); Schwartz, Caprara, and Vecchione (2010); Schwartz et al. (2014); Seewann (2022)
International cooperation (e.g., EU membership)	Mixed	Mixed	Dennison, Seddig, and Davidov (2021); Goren et al. (2016); Rathbun et al. (2016); Reinl et al. (2024)
Law, order, and civil liberties (e.g., limiting terrorism by restricting movement)	Left	Right	Schwartz, Caprara, and Vecchione (2010); Schwartz et al. (2014)
Military interventions (e.g., defense spending)	Left	Right	Crawford, Wiley, and Ventresco (2014); Goren et al. (2016); O'Dwyer and Çoymak (2020); Rathbun et al. (2016); Schwartz, Caprara, and Vecchione (2010); Schwartz et al. (2014)
Nuclear power	Left	Right	Perlaviciute and Steg (2015); Whitfield et al. (2009)
Racial equality (e.g., aid to Black people)	Left	Right	Goren et al. (2016); Grigoryan and Schwartz (2021)
Social mores (e.g., gay marriage)	Mixed	Right	Abdallah (2023); Dobewall and Rudnev (2014); Donaldson, Handren, and Lac (2017); Goren et al. (2016); Kuntz et al. (2015); Schwartz, Caprara, and Vecchione (2010); Schwartz (2012); Schwartz et al. (2014); Wu and Molden (2024)

To identify which policy items in the panels should relate to conservation and self-transcendence, I conducted a systematic literature review before the main analyses. On February 10, 2026, I searched Scopus for English-language journal articles published since 2010. Their titles, abstracts, or keywords had to match *values AND (politic* OR polic*) AND (attitud* OR belief* OR opinion* OR preference*)*. The full text also had to name at least one relevant Schwartz value (*self-transcendence, conservation, universalism, benevolence, conformity, tradition, or security*) and cite an author named Schwartz.

I screened abstracts, checking full texts when eligibility was ambiguous, for articles that quantitatively estimated the relationship between basic values and policy opinions in a broad sample. This yielded 24 articles. I added 18 articles known to me or located through a similar Google Scholar search. I reviewed all 42 full texts, recording the direction and significance of each value-policy association and the policy type it concerned.

When studies reported both higher-order values (such as conservation) and their lower-order components (such as tradition and security), I used the higher-order results. When only lower-order results were available, I inferred the higher-order association if the significant effects did not conflict. For example, consistently positive effects, with or without null effects for other components, implied a positive conservation association; significant effects in opposite directions were coded as mixed.

I deemed an association consistent across studies and respondent subgroups only if every significant estimate pointed in the same direction. Null effects did not count against consistency because they could reflect limited power; significant effects in opposite directions ruled it out. Table B1 reports the resulting map and its sources. The “S.T.” and “CON.” columns show whether each value generally predicts support for left-wing, right-wing, or mixed positions in a policy area.

Appendix C: Robustness Checks

This appendix reports nine robustness checks on the main results. Each re-estimates the main models after changing one element of the design: which policy items enter the analysis, how value-consistency is coded, how opinions are measured, or which panelists are analyzed. Table C1 collects the resulting stability gaps, one block of rows per check (three for the sophistication check), beside each outcome’s Cohen’s $d = 0.2$ benchmark. The continuous-measure check, whose estimates are not percentage-point gaps, is reported only in its subsection below.

C.1 Results When Including Shorter Scales

The main analyses use only five- to seven-point Likert items for comparability. Here, I extended the analysis to all relevant policy items, including binary ones. The all-items rows of Table C1 show that the gaps closely mirror those in the main paper.

C.2 Results With Empirical Alignment

The main analyses rely on a literature-derived mapping of values to policy opinions, which may involve contestable judgment calls. As an alternative, I replicated the analyses using a purely empirical approach. For each policy opinion, I estimated the following regression:

$$Opinion_{i,t} = \beta_1 * SelfTranscendence_i + \beta_2 * Conservation_i + \beta_3 * Acquiescence_i + \delta_t + \epsilon$$

Here, *Opinion* is panelist i ’s position on the issue at wave t , *SelfTranscendence* and *Conservation* capture endorsement of each value dimension, *Acquiescence* controls for indiscriminate agreement with value statements, and δ_t represents wave fixed effects. I adjusted for multiple comparisons using the Holm (1979) method ($p < 0.05$). If a value had a significant effect on a policy opinion, after adjusting for multiple comparisons, the direction of that effect dictated the “proper” direction of the association.

Table C1: Stability Gaps Across Robustness Checks

Panel	Same Opinion		Same Side	
	Gap (SE) [95% CI]	Cohen's $d = 0.2$	Gap (SE) [95% CI]	Cohen's $d = 0.2$
All Policy Items				
Germany	3.3 (0.2) [2.9, 3.7]	9.8	4.1 (0.2) [3.8, 4.4]	9.0
Netherlands	1.9 (0.4) [1.1, 2.7]	8.8	3.0 (0.4) [2.2, 3.8]	7.7
United States	4.2 (0.6) [2.9, 5.4]	8.5	4.3 (0.6) [3.0, 5.6]	8.3
Empirical Alignment				
Germany	2.6 (0.2) [2.2, 3.1]	9.8	3.5 (0.2) [3.1, 3.9]	9.2
Netherlands	1.8 (0.5) [0.9, 2.7]	9.7	2.7 (0.5) [1.8, 3.5]	8.5
United States	9.9 (3.9) [2.3, 17.4]	9.7	7.6 (3.6) [0.5, 14.7]	8.9
Middle-Aged Panelists				
Germany	3.0 (0.3) [2.5, 3.5]	9.7	3.7 (0.2) [3.2, 4.1]	8.9
Netherlands	1.7 (0.7) [0.3, 3.1]	9.4	4.0 (0.7) [2.7, 5.4]	8.0
United States	4.9 (2.9) [-0.7, 10.6]	9.6	1.9 (2.7) [-3.5, 7.2]	8.9
Policy-Opinion Indexes				
Netherlands	1.5 (0.5) [0.6, 2.4]	9.5	4.2 (0.5) [3.3, 5.2]	8.3
United States	4.9 (1.5) [1.9, 7.9]	8.3	4.9 (1.6) [1.8, 8.0]	8.1
Values Most Important				
Germany	7.1 (0.4) [6.2, 8.0]	9.7	7.9 (0.4) [7.1, 8.7]	8.8
Netherlands	3.6 (1.0) [1.6, 5.5]	9.5	5.6 (1.0) [3.7, 7.6]	8.2
United States	7.5 (4.3) [-0.9, 16.0]	9.8	10.0 (3.7) [2.6, 17.3]	8.8
Strong Value-Policy Cases				
Germany	6.2 (0.4) [5.5, 6.9]	9.7	7.7 (0.3) [7.0, 8.3]	8.8
Netherlands	2.9 (0.8) [1.4, 4.4]	9.7	5.8 (0.7) [4.3, 7.2]	8.7
United States	9.9 (3.9) [2.3, 17.4]	9.7	7.6 (3.6) [0.5, 14.7]	8.9
Most Educated Panelists				
Germany	3.2 (0.3) [2.6, 3.9]	9.8	4.8 (0.3) [4.2, 5.4]	8.8
Netherlands	2.3 (0.8) [0.8, 3.8]	9.4	3.8 (0.8) [2.4, 5.3]	7.9
United States	7.3 (3.2) [1.2, 13.5]	9.7	8.5 (2.9) [2.8, 14.2]	8.9
Most Politically Interested Panelists				
Germany	4.3 (0.4) [3.5, 5.1]	9.7	5.2 (0.4) [4.4, 5.9]	8.7
Netherlands	3.4 (0.7) [2.1, 4.7]	9.4	4.7 (0.6) [3.5, 5.8]	8.0
United States	3.4 (3.3) [-3.0, 9.8]	9.6	4.7 (3.3) [-1.9, 11.3]	9.0
Most Politically Knowledgeable Panelists				
Germany	3.6 (0.3) [3.0, 4.2]	9.7	4.7 (0.3) [4.2, 5.3]	8.6
Issues Implicating Both Values				
Germany	6.4 (0.4) [5.7, 7.1]	9.8	6.6 (0.3) [6.0, 7.3]	8.9
Netherlands	4.1 (1.1) [2.0, 6.2]	9.7	7.5 (1.1) [5.4, 9.6]	8.7
United States	0.3 (3.6) [-6.8, 7.4]	9.8	-1.1 (3.3) [-7.6, 5.4]	9.2

Note: Gaps and Cohen's $d = 0.2$ cutoffs are in percentage points. Parentheses give wild-bootstrap standard errors clustered by panelist, and brackets give 95% confidence intervals from them. Each cutoff equals 0.2 times the weighted, demeaned standard deviation of its outcome. Data are weighted to give each panelist and, as nearly as the data allow, each value-policy pair equal weight.

The empirical-alignment rows of Table C1 present these results, which broadly match the main findings. The U.S. estimates grow noisier under this specification but remain consistent with a small effect. In the U.S. panel, only three of the Holm-significant pairs have five or more response options (immigration levels, same-sex marriage, and taxes on high incomes), and the strong-case check retains the same three, so the two blocks' U.S. rows coincide.

C.3 Results Among the Middle-Aged

If over-time value change were attenuating the estimates from my main analyses, restricting the sample to panelists with the most stable values should alter the results. Past research identifies 40- to 60-year-olds as the group least likely to shift their values (Milfont, Milojev, and Sibley 2016). The middle-aged rows of Table C1 show that the gaps track the full-sample results, though the U.S. estimates are imprecise.

C.4 Results Using Indexes of Policy Opinions

Measurement error in individual policy items could inflate instability. To guard against this, I replicated the analyses using policy-opinion indexes (Ansolabehere, Rodden, and Snyder 2008). Within the Dutch and U.S. panels, I identified pairs of policy opinions that were measured in at least two of the same waves, conceptually related, and strongly correlated ($r < -0.3$ or $r > 0.3$). This process yielded 49 indexes. Because each index averages its two items, I kept only observations in which a panelist answered both items at both waves of a period. I omitted the German panel because its paired items were rarely asked in the same waves: Only about 5% of German observations would include both items. The policy-index rows of Table C1 show that the gaps remain modest.

C.5 Results When Using Continuous Measures

The main analyses rely on binary indicators of value-consistency and opinion stability. Binarization simplifies interpretation but may introduce measurement error, so I re-estimated the models using continuous alternatives. Value-consistency is measured as one minus the absolute distance between a panelist’s value score and policy opinion (flipped so that they correlate positively). Opinion stability is similarly defined as one minus the absolute change in the panelist’s opinion across waves.

The value-consistency coefficients are 0.027 (SE 0.002) in Germany, 0.054 (0.005) in the Netherlands, and 0.056 (0.016) in the United States. The link remains modest. The U.S. coefficient implies that moving from the bottom to the top of the value-consistency scale cuts the average change in policy opinions by six percentage points. Put differently, a one-standard-deviation increase in value-consistency increases opinion stability by roughly 5% of a standard deviation. The full models are in Replication Materials, Table S4.

C.6 Results When Values Are Most Important

Conservation and self-transcendence may matter little to some respondents, potentially diluting the consistency-stability link. To address this, I first computed value importance following Schwartz et al. (2001): For each participant, I calculated the absolute distance between their self-transcendence and conservation ratings and their mean agreement across all values. I then re-ran the main models restricted to the top third of respondents on this importance measure within each value-policy pairing. The value-importance rows of Table C1 show that the gaps remain small relative to the Cohen benchmark. The U.S. estimates are imprecise, and the same-side gap slightly exceeds its cutoff.

C.7 Results for Strongest Value-Policy Cases

Perhaps the modest consistency-stability link reflects the distance between basic values and politics. For each issue and dataset, I regressed the three-point policy opinion on its corresponding value, acquiescence (Schwartz 1992), and wave fixed effects. Within each panel, I retained the third of pairs with the strongest standardized value coefficients in the expected direction and re-estimated the stability models. The selection kept 3 of 9 pairs in the United States (coefficients of 0.20–0.42), 5 of 15 in the Netherlands (0.23–0.31), and 15 of 43 in Germany (0.29–0.44). The strong-case rows of Table C1 show that the gaps remain statistically lower than, or indistinguishable from, a small effect. The U.S. estimates are imprecise.

C.8 Results Among Sophisticated Panelists

I re-estimated the main models among panelists in the top third of education and political interest within each panel and, in the German panel, among panelists in the top third of a factual political-knowledge index. Because these measures take discrete values, respondents who shared the value at each cutoff were kept or dropped together, whichever brought the subset nearer one-third of observations (17–39%). Only the German panel measures political knowledge directly: The Dutch panel contains no factual political-knowledge items, and the U.S. panel’s only knowledge measure is a vocabulary test without political content. Education and political interest therefore serve as proxies in all three panels (Delli Carpini and Keeter 1996); values predict policy opinions more strongly among the more educated and politically interested (Replication Materials, Figure S1). The German index draws on 16 items: the electoral system (the 5% threshold, the first and second votes, and the election of the Chancellor), the current unemployment rate, and the party affiliations of 12 prominent politicians. At each wave, I averaged all the items a panelist had answered so far, using the most recent response to each and scoring “don’t know” as incorrect, into an index from 0 to 1 ($n \approx 27,000$ panelists also have value measures). Because the items were asked in different

waves, the number available varies: Panelists recruited in 2020, for instance, answered only the electoral-system items. Among panelists who answered more, the index correlates at $r = 0.70$ with a score built from the electoral-system items alone. The three sophistication blocks of Table C1 report the resulting stability gaps. The gaps remain small across all three groups, although the U.S. estimates are less precise.

C.9 Results When Issues Implicate Both Values

The main analyses code value-consistency against one value at a time, so an opinion coded as inconsistent with, say, self-transcendence may still advance the respondent’s conservation values. To address this, I re-estimated the models on the subset of policy issues that the literature ties to both self-transcendence and conservation values (see Table B1 in app. B). For these issues, the two values push in opposite directions: Self-transcendence predicts the left-wing position, and conservation predicts the right-wing position. I therefore identified which of the two values each panelist prioritized—using the difference between their conservation and self-transcendence scores—and coded an opinion as value-consistent when it fell on the side of the issue associated with the prioritized value. Each issue entered the analysis once, and panelists who endorsed both values equally were excluded. I then re-estimated the main models with this recoded consistency measure, retaining the same covariates, fixed effects, weights, and bootstrapped standard errors.

The both-values rows of Table C1 show that accounting for both values simultaneously does not change my conclusions. The consistency-stability gap is indistinguishable from zero in the United States. While statistically significant in Germany and the Netherlands, this gap remains below the Cohen’s $d = 0.2$ benchmark in every panel—and far below Converse’s expectation.

Appendix D: Modeling Case-Wise Heterogeneity

To assess whether the association between value-consistency and stability varies by value or policy issue, I estimated a separate OLS regression for each value-policy pairing:

$$OpinionStable_{i,t} = \beta_1 * ValueConsistent_{i,t} + \beta * Traits_{i,t} + \delta_{s,t} + \epsilon$$

Here, *OpinionStable* equals one if panelist *i*’s opinion remained on the same side of the issue over period *t*, and *ValueConsistent* equals one if the opinion aligned with the relevant value at the period’s start. *Traits* includes education, political interest, age, and—following convention in values research—acquiescence (Schwartz 1992). “Trend” fixed effects (δ) absorb variation due to the initial stance held on a policy issue (e.g., left, center, or right). Figure 3 in the main paper visualizes the results of these models.

Appendix E: Does Time Affect the Stability Gap?

Longer gaps between waves might erode opinion stability or widen the gap between value-consistent and -inconsistent opinions. To test this, I estimated the following OLS regression for each panel:

$$\begin{aligned} OpinionStable_{i,p,t} = & \beta_1 * ValueConsistent_{i,p,v,t} + \beta_2 * YearsElapsed_{i,p,t} \\ & + \beta_3 * ValueConsistent_{i,p,v,t} * YearsElapsed_{i,p,t} + \alpha_{p,v} + \delta_{s,p} + \epsilon \end{aligned}$$

The dependent variable, *OpinionStable*, equals one if panelist *i* held the same position on policy issue *p* across period *t*. *ValueConsistent* flags whether that opinion aligned with value *v* at the period’s start, and *YearsElapsed* measures the period’s duration in years.

Two sets of fixed effects address unobserved heterogeneity. “Case” fixed effects (α) restrict comparisons to within a given value-policy pairing. “Trend” fixed effects (δ) absorb differences tied to the initial stance on an issue (e.g., left, center, or right), since some positions—such as moderate ones—may be inherently less stable.

Table E1: Does Period Length Affect the Stability Gap?

	Germany		Netherlands		United States	
	Same Opinion	Same Side	Same Opinion	Same Side	Same Opinion	Same Side
Value-Consistent	0.0319 (0.0023)	0.0364 (0.0021)	0.0253 (0.0075)	0.0371 (0.0068)	0.0343 (0.0325)	0.0539 (0.0258)
Years Between Waves	-0.0220 (0.0012)	-0.0218 (0.0012)	-0.0069 (0.0010)	-0.0048 (0.0009)	-0.0202 (0.0080)	-0.0113 (0.0077)
Interaction	0.0021 (0.0014)	0.0065 (0.0013)	0.0008 (0.0013)	0.0007 (0.0013)	0.0046 (0.0108)	-0.0011 (0.0081)
Observations	10,190,569	10,190,569	768,551	768,551	11,084	11,084
RMSE	0.49	0.45	0.48	0.41	0.49	0.46
Cohen's $d = 0.2$	0.0977	0.0899	0.0954	0.0827	0.0977	0.0913

Note: Each model includes case and trend fixed effects. Data weighted to give each panelist and, as nearly as the data allow, each value-policy pair equal weight. Wild-bootstrap standard errors clustered by panelist in parentheses. The Cohen row gives the main-model small-effect reference for level gaps; the interaction measures change in that gap per year.

As shown in Table E1, longer intervals between waves reduce opinion stability, significantly so in five of the six models. Yet they barely widen the consistency-stability gap. The interaction is statistically significant only for same-side stability in Germany, where the gap grows by 0.7 percentage points per year. Even at the longest interval in that panel (5.2 years), the implied gap of 7.0 percentage points remains below the Cohen's $d = 0.2$ benchmark.

Appendix F: Stability of Political Values

Scholars question whether political values are distinct from, or exogenous to, policy opinions (Arceneaux et al. 2025). Rather than shaping policy opinions, these values may simply summarize citizens’ views within a particular domain.

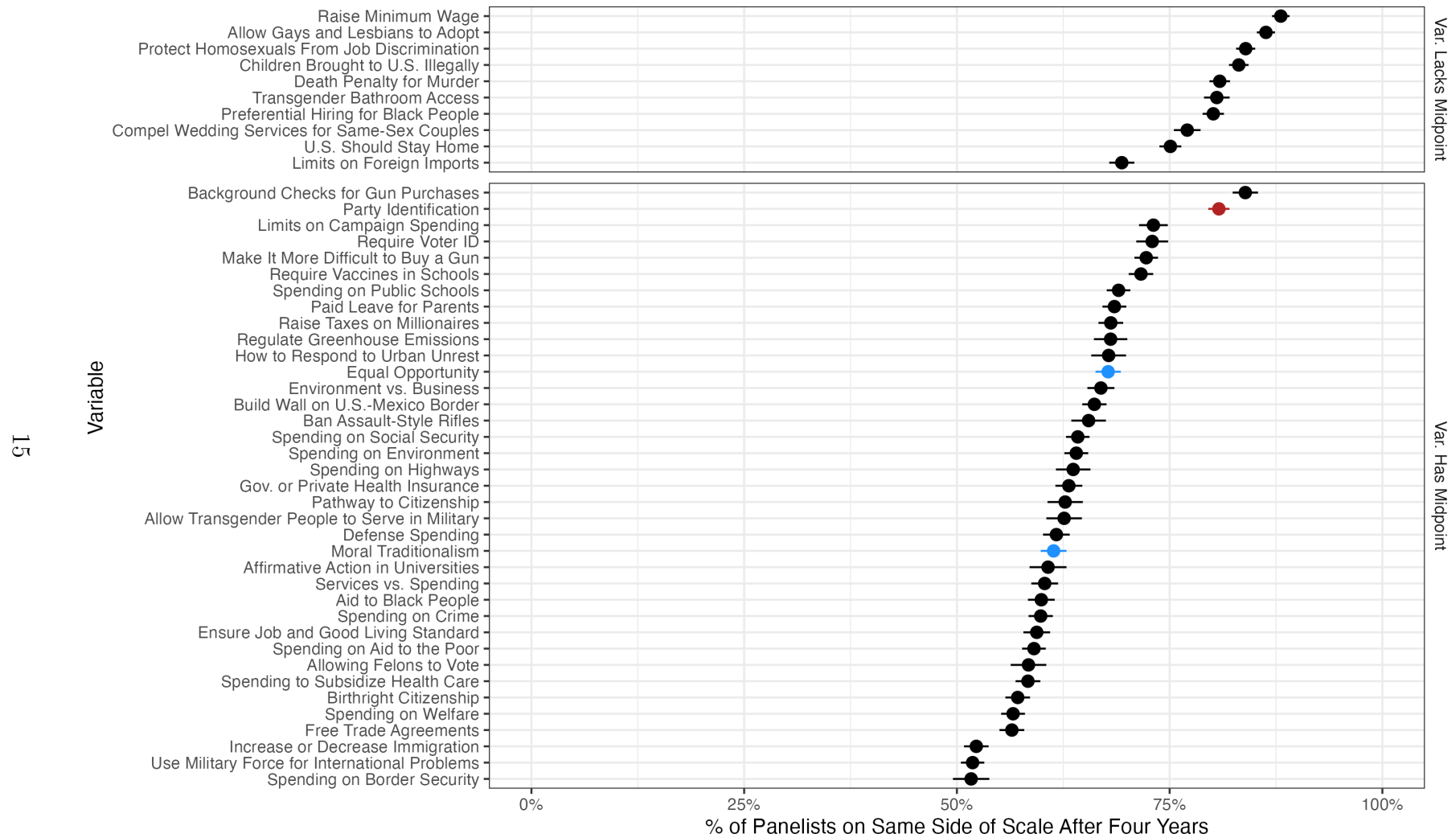
To investigate, I drew on the 1992–96 and 2016–24 panels from the American National Election Studies (ANES; American National Election Studies 1999, 2025). All variables were coded as left-wing, scale midpoint (if any), or right-wing to ensure comparability, and the data were restructured so each row represents panelist i ’s score on variable v at the beginning and end of period t . I then estimated this OLS regression for each dataset:

$$Stable_{i,v,t} = \beta * Variable_v + \delta_t + \epsilon$$

Stable equals one if panelist i stood on the same side of variable v across period t ; *Variable* is a factor identifying the variable; and δ captures period fixed effects.

Figure F1 displays predicted stability in the 2016–24 panel. The 1992–96 panel shows the same broad pattern: Political values are roughly as unstable as policy opinions (full plot in Replication Materials, Figure S2).

Figure F1: Stability of Political Values Versus Policy Opinions (ANES 2016–24 Panel)



Note: Predicted stability of variables in the ANES 2016–24 panel. Blue points represent political value indexes, black points represent policy opinions, and red points represent party identification. Lines represent 95% confidence intervals from standard errors clustered by panelist.

Appendix G: Stability of Basic Values

A separate German panel—the Leibniz Institute for the Social Sciences (GESIS) panel—allows a direct comparison of the stability of basic values and policy opinions (GESIS 2025). It measures 6,165 panelists’ basic values across 11 waves and also contains repeated measures of 44 policy opinions.

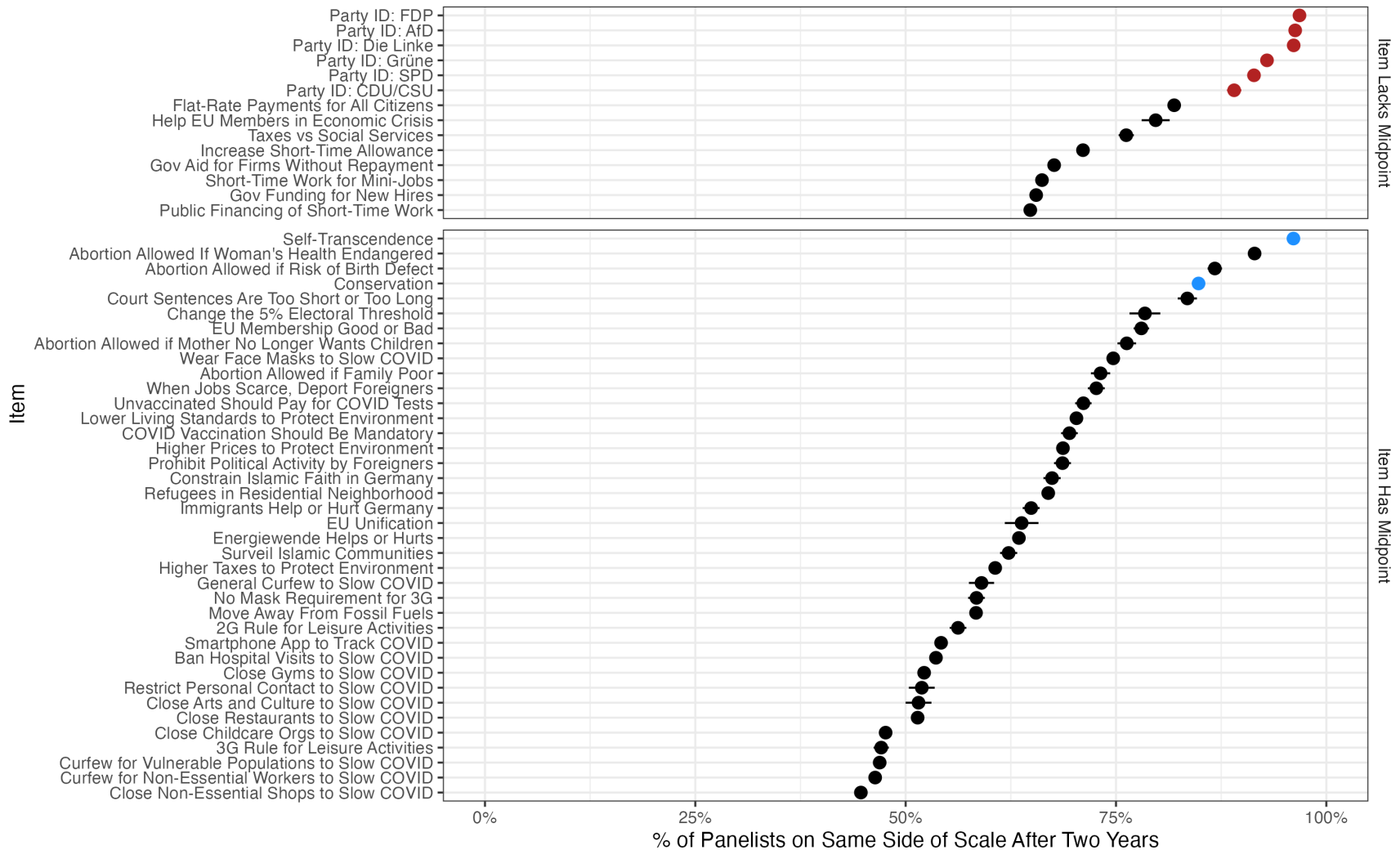
As in appendix F, all variables were coded as left-wing, scale midpoint (if any), or right-wing for comparability, and the data were restructured so each row represents panelist i ’s score on variable v at the beginning and end of period t . I then estimated:

$$Stable_{i,v,t} = \beta_0 + \beta_1 * Years_t + \beta * Variable_v + \epsilon$$

Stable is defined as before; *Years* captures the period length; and *Variable* identifies the variable. I did not use period fixed effects in this model because of insufficient overlap in variables across waves.

Figure G1 displays predicted stability after two years. After one year, 97% and 86% of panelists remained stable on self-transcendence and conservation values, respectively. Most policy opinions were far less stable (average stability = 66%). Only two policy opinions showed more stability than conservation values. The 2G and 3G items in Figure G1 refer to COVID-19 rules that restricted entry to people who were vaccinated or recovered (2G) or who were vaccinated, recovered, or tested (3G).

Figure G1: Stability of Basic Values Versus Policy Opinions



Note: Predicted stability of variables in the GESIS panel. Blue points represent basic value indexes, black points represent policy opinions, and red points represent party identification. Lines represent 95% confidence intervals from standard errors clustered by panelist.

Appendix H: Value-Policy Correlations Over Time

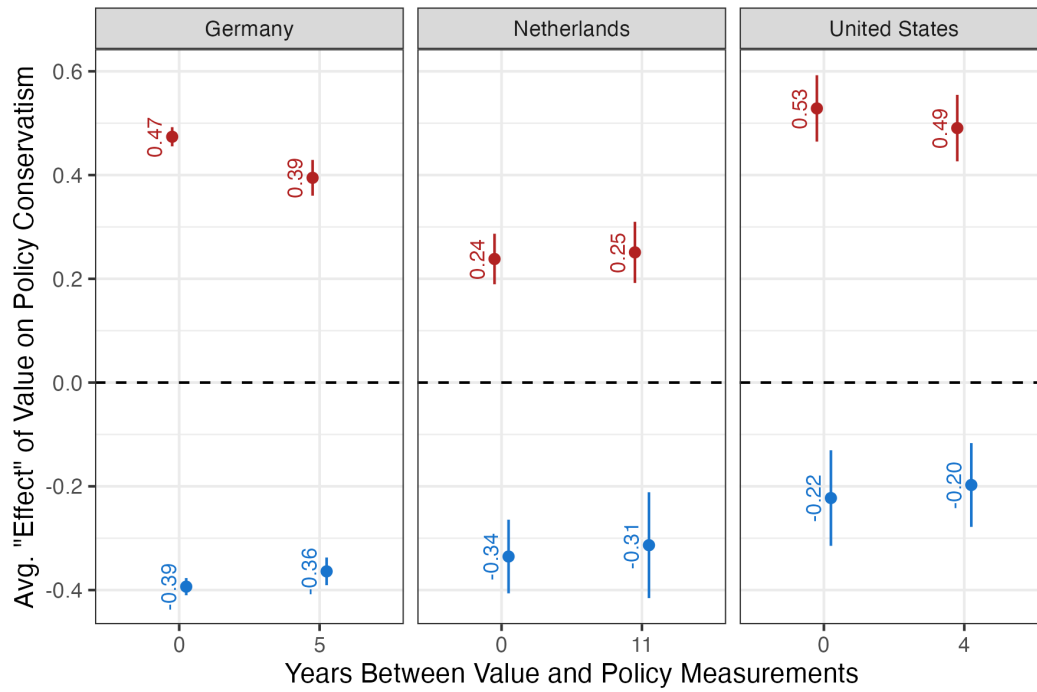
Each panel in the main analysis measures values once—at the panelist’s first wave (Germany), the panel’s start (Netherlands), or its end (United States)—so I cannot directly verify that values remained stable throughout. I can, however, test an implication of value change: If values shifted, their average cross-sectional association with related policy opinions should weaken as the measurement grows stale. Individual value-policy correlations may increase or decrease with changing circumstances, but an average across many opinions should predominantly reflect the passage of time. For each value and panel, I estimated:

$$\begin{aligned} Policy_{i,p,t} = & \beta_1 * Value_i + \beta_2 * Years_{i,p,t} + \beta_3 * Acquiescence_i \\ & + \beta_4 * Value_i * Years_{i,p,t} + \beta_5 * Years_{i,p,t} * Acquiescence_i + \beta_6 * Value_i * Acquiescence_i \\ & + \beta_7 * Value_i * Years_{i,p,t} * Acquiescence_i + \gamma_p + \epsilon_{i,p,t} \end{aligned}$$

Policy is panelist *i*’s opinion on issue *p* at wave *t*; *Value* captures endorsement of the related value; *Years* is the gap (in years) between the value and policy measurements; *Acquiescence* controls for indiscriminate agreement with value statements; and γ denotes policy-issue fixed effects. I used a complete-interaction model to minimize bias in my estimates (Blackwell and Olson 2022).

Figure H1 displays the associational “effect” of panelists’ values on their policy opinions when values were measured and at the longest gap in each panel: 4 years in the United States, 5 in Germany, and 11 in the Netherlands. These effects change little, though each panel spans many years. Five of the six weaken and one strengthens, but even the largest shift, for conservation values in Germany, is 0.08, about a sixth of the effect at measurement. In the United States and the Netherlands, each estimate at the longest gap lies within the confidence interval of its counterpart at measurement. In Germany, whose large sample yields narrow intervals, neither does. Standard errors grow most with the gap in Germany, where fewer panelists were observed long after their values were measured. If true value change

Figure H1: Associational “Effect” of Values on Policy Conservatism by Time



Note: This figure shows the associational “effect” of panelists’ values on policy conservatism when values were measured and at the longest gap between value and policy measurements in each panel. Blue points represent the effects of self-transcendence values, while red points represent the effects of conservation values. Data are weighted to give each panelist and, as nearly as the data allow, each value-policy pair equal weight. Lines represent 95% confidence intervals from wild-bootstrap standard errors clustered by panelist.

were occurring, we would expect the coefficients to decay toward zero. Instead, each retains at least four-fifths of its size at measurement.

References

- Abdallah, Saamah. 2023. “The Relative Importance of Self-Focused and Society-Focused Values in Explaining Political Attitudes in Europe.” *Political Psychology* 44 (3): 455–473.
- Ahmadinia, Hamed. 2025. “Shaping Immigration Attitudes: The Role of Human Values, Media Engagement and Sociopolitical Events among European Managers.” *Nordic Journal of Migration Research* 16 (1): 6.
- American National Election Studies. 1999. “ANES 1992–1997 Merged File [Dataset].” Ann Arbor, MI: University of Michigan. <https://electionstudies.org/data-center/1992-1997-merged-file/> (accessed October 5, 2026).
- American National Election Studies. 2025. “ANES 2016–2020–2024 Panel Study Merged File (October 30, 2025 Version) [Dataset].” Ann Arbor, MI: University of Michigan. <https://electionstudies.org/data-center/2016-2020-2024-panel-merged-study/> (accessed October 5, 2026).
- Ansolabehere, Stephen, Jonathan Rodden, and James M. Snyder. 2008. “The Strength of Issues: Using Multiple Measures to Gauge Preference Stability, Ideological Constraint, and Issue Voting.” *American Political Science Review* 102 (2): 215–232.
- Arceneaux, Kevin, Bert N. Bakker, Neil Fasching, and Yphtach Lelkes. 2025. “A Critical Evaluation and Research Agenda for the Study of Psychological Dispositions and Political Attitudes.” *Political Psychology* 46 (S1): 7–37.
- Arikan, Gizem, and Eser Sekercioglu. 2019. “Authoritarian Predispositions and Attitudes towards Redistribution.” *Political Psychology* 40 (5): 1099–1118.
- Bergquist, Magnus, Andreas Nilsson, Niklas Harring, and Sverker C. Jagers. 2022. “Meta-analyses of Fifteen Determinants of Public Opinion about Climate Change Taxes and Laws.” *Nature Climate Change* 12 (3): 235–240.
- Blackwell, Matthew, and Michael P. Olson. 2022. “Reducing Model Misspecification and Bias in the Estimation of Interactions.” *Political Analysis* 30 (4): 495–514.

- Cichocki, Piotr, and Piotr Jabkowski. 2019. "Immigration Attitudes in the Wake of the 2015 Migration Crisis in the Visegrád Group Countries: Comparative Insights of ESS7 and ESS8." *Intersections: East European Journal of Society and Politics* 5 (1): 27–47.
- Crawford, Jarret T., Shaun Wiley, and Nina Ventresco. 2014. "Examining Americans' Attitudes toward Drone Strikes on the Eve of the 2012 Presidential Election." *Analyses of Social Issues and Public Policy* 14 (1): 46–60.
- Davidov, Eldad, and Bart Meuleman. 2012. "Explaining Attitudes towards Immigration Policies in European Countries: The Role of Human Values." *Journal of Ethnic and Migration Studies* 38 (5): 757–775.
- Davidov, Eldad, Bart Meuleman, Jaak Billiet, and Peter Schmidt. 2008. "Values and Support for Immigration: A Cross-Country Comparison." *European Sociological Review* 24 (5): 583–599.
- Delli Carpini, Michael X., and Scott Keeter. 1996. *What Americans Know about Politics and Why It Matters*. New Haven, CT: Yale University Press.
- Dennison, James, Eldad Davidov, and Daniel Seddig. 2020. "Explaining Voting in the UK's 2016 EU Referendum: Values, Attitudes to Immigration, European Identity and Political Trust." *Social Science Research* 92:102476.
- Dennison, James, Daniel Seddig, and Eldad Davidov. 2021. "The Role of Human Values in Explaining Support for European Union Membership." *Journal of Cross-Cultural Psychology* 52 (4): 372–387.
- Dobewall, Henrik, and Maksim Rudnev. 2014. "Common and Unique Features of Schwartz's and Inglehart's Value Theories at the Country and Individual Levels." *Cross-Cultural Research* 48 (1): 45–77.
- Donaldson, Candice D., Lindsay M. Handren, and Andrew Lac. 2017. "Applying Multilevel Modeling to Understand Individual and Cross-Cultural Variations in Attitudes toward Homosexual People across 28 European Countries." *Journal of Cross-Cultural Psychology* 48 (1): 93–112.

- GESIS. 2025. “GESIS Panel – Standard Edition (ZA5665; Version 56.0.0) [Dataset].” Cologne: GESIS – Leibniz Institute for the Social Sciences. <https://doi.org/10.4232/1.14491> (accessed October 5, 2026).
- Goren, Paul, Harald Schoen, Jason Reifler, Thomas Scotto, and William Chittick. 2016. “A Unified Theory of Value-Based Reasoning and U.S. Public Opinion.” *Political Behavior* 38 (4): 977–997.
- Grigoryan, Lusine, and Shalom H. Schwartz. 2021. “Values and Attitudes towards Cultural Diversity: Exploring Alternative Moderators of the Value–Attitude Link.” *Group Processes & Intergroup Relations* 24 (6): 966–981.
- Grunert, Suzanne C., and Hans Jørn Juhl. 1995. “Values, Environmental Attitudes, and Buying of Organic Foods.” *Journal of Economic Psychology* 16 (1): 39–62.
- Holm, Sture. 1979. “A Simple Sequentially Rejective Multiple Test Procedure.” *Scandinavian Journal of Statistics* 6 (2): 65–70.
- Kuntz, Anabel, Eldad Davidov, Shalom H. Schwartz, and Peter Schmidt. 2015. “Human Values, Legal Regulation, and Approval of Homosexuality in Europe: A Cross-Country Comparison.” *European Journal of Social Psychology* 45 (1): 120–134.
- Messing, Vera, and Bence Ságvári. 2021. “Are Anti-immigrant Attitudes the Holy Grail of Populists? A Comparative Analysis of Attitudes towards Immigrants, Values, and Political Populism in Europe.” *Intersections: East European Journal of Society and Politics* 7 (2): 100–127.
- Miglietta, Anna, Stefano Tartaglia, and Barbara Loera. 2018. “Personal Values and the Acceptance of Immigrants: Why National Identification Matters.” *International Journal of Social Psychology* 33 (3): 682–714.
- Milfont, Taciano L., and Valdiney V. Gouveia. 2006. “Time Perspective and Values: An Exploratory Study of Their Relations to Environmental Attitudes.” *Journal of Environmental Psychology* 26 (1): 72–82.

- Milfont, Taciano L., Petar Milojev, and Chris G. Sibley. 2016. "Values Stability and Change in Adulthood: A 3-Year Longitudinal Study of Rank-Order Stability and Mean-Level Differences." *Personality and Social Psychology Bulletin* 42 (5): 572–588.
- Nilsson, Andreas, Chris von Borgstede, and Anders Biel. 2004. "Willingness to Accept Climate Change Strategies: The Effect of Values and Norms." *Journal of Environmental Psychology* 24 (3): 267–277.
- O'Dwyer, Emma, and Ahmet Çoymak. 2020. "Basic Human Values and Their Contexts: A Multilevel Analysis of Support for the Use of Armed Drones in the United States, United Kingdom, and Turkey." *Political Psychology* 41 (2): 249–264.
- Oh, Soomin, David Hudson, and Jennifer Hudson. 2025. "Public Preferences on the Economy–Environment Trade-Off in France, Germany, Great Britain, and the United States." *Politics & Policy* 53 (3): e70047.
- Perlavičiute, Goda, and Linda Steg. 2015. "The Influence of Values on Evaluations of Energy Alternatives." *Renewable Energy* 77:259–267.
- Rathbun, Brian C., Joshua D. Kertzer, Jason Reifler, Paul Goren, and Thomas J. Scotto. 2016. "Taking Foreign Policy Personally: Personal Values and Foreign Policy Attitudes." *International Studies Quarterly* 60 (1): 124–137.
- Reinl, Ann-Kathrin, Daniel Seddig, James Dennison, and Eldad Davidov. 2024. "Basic Human Values and Preferences for an EU-Wide Social Benefit Scheme." *JCMS: Journal of Common Market Studies* 62 (2): 564–582.
- Schwartz, Shalom H. 1992. "Universals in the Content and Structure of Values: Theoretical Advances and Empirical Tests in 20 Countries." In Mark P. Zanna, ed., *Advances in Experimental Social Psychology*. Vol. 25. San Diego, CA: Academic Press, 1–65.
- Schwartz, Shalom H. 2007a. "Universalism Values and the Inclusiveness of Our Moral Universe." *Journal of Cross-Cultural Psychology* 38 (6): 711–728.
- Schwartz, Shalom H. 2007b. "Value Orientations: Measurement, Antecedents and Consequences across Nations." In Roger Jowell, Caroline Roberts, Rory Fitzgerald, and Gillian

- Eva, eds., *Measuring Attitudes Cross-Nationally: Lessons from the European Social Survey*. London: SAGE, 169–203.
- Schwartz, Shalom H. 2012. “An Overview of the Schwartz Theory of Basic Values.” *Online Readings in Psychology and Culture* 2 (1). <https://doi.org/10.9707/2307-0919.1116>.
- Schwartz, Shalom H., Gian Vittorio Caprara, and Michele Vecchione. 2010. “Basic Personal Values, Core Political Values, and Voting: A Longitudinal Analysis.” *Political Psychology* 31 (3): 421–452.
- Schwartz, Shalom H., Gian Vittorio Caprara, Michele Vecchione, Paul Bain, Gabriel Bianchi, Maria Giovanna Caprara, Jan Cieciuch, Hasan Kirmanoglu, Cem Baslevent, Jan-Erik Lönnqvist, Catalin Mamali, Jorge Manzi, Vassilis Pavlopoulos, Tetyana Posnova, Harald Schoen, Jo Silvester, Carmen Tabernero, Claudio Torres, Markku Verkasalo, Eva Vondráková, Christian Welzel, and Zbigniew Zaleski. 2014. “Basic Personal Values Underlie and Give Coherence to Political Values: A Cross National Study in 15 Countries.” *Political Behavior* 36 (4): 899–930.
- Schwartz, Shalom H., Gila Melech, Arielle Lehmann, Steven Burgess, Mari Harris, and Vicki Owens. 2001. “Extending the Cross-Cultural Validity of the Theory of Basic Human Values with a Different Method of Measurement.” *Journal of Cross-Cultural Psychology* 32 (5): 519–542.
- Seewann, Lena. 2022. “My Values, Their Values: How Value Conceptualisations Influence Attitudes towards Immigration.” *Journal of Ethnic and Migration Studies* 48 (9): 2091–2114.
- Smith, E. Keith, and Lynn M. Hempel. 2022. “Alignment of Values and Political Orientations Amplifies Climate Change Attitudes and Behaviors.” *Climatic Change* 172 (1–2): 4.
- Steg, Linda, Judith I. M. De Groot, Lieke Dreijerink, Wokje Abrahamse, and Frans Siero. 2011. “General Antecedents of Personal Norms, Policy Acceptability, and Intentions: The Role of Values, Worldviews, and Environmental Concern.” *Society & Natural Resources* 24 (4): 349–367.

- Steg, Linda, Lieke Dreijerink, and Wokje Abrahamse. 2005. "Factors Influencing the Acceptability of Energy Policies: A Test of VBN Theory." *Journal of Environmental Psychology* 25 (4): 415–425.
- Stern, Paul C., Thomas Dietz, and Gregory A. Guagnano. 1998. "A Brief Inventory of Values." *Educational and Psychological Measurement* 58 (6): 984–1001.
- Stern, Paul C., Linda Kalof, Thomas Dietz, and Gregory A. Guagnano. 1995. "Values, Beliefs, and Proenvironmental Action: Attitude Formation toward Emergent Attitude Objects." *Journal of Applied Social Psychology* 25 (18): 1611–1636.
- Whitfield, Stephen C., Eugene A. Rosa, Amy Dan, and Thomas Dietz. 2009. "The Future of Nuclear Power: Value Orientations and Risk Perception." *Risk Analysis* 29 (3): 425–437.
- Wu, Eileen Z., and Daniel C. Molden. 2024. "The Moderating Role of Democratic Governance in the Association between Personal Values and Political Ideologies." *Political Psychology* 45 (3): 601–622.